

Supported Platinum Nanoparticles Catalyzed Carbon–Carbon Bond Cleavage of Polyolefins: Role of the Oxide Support Acidity

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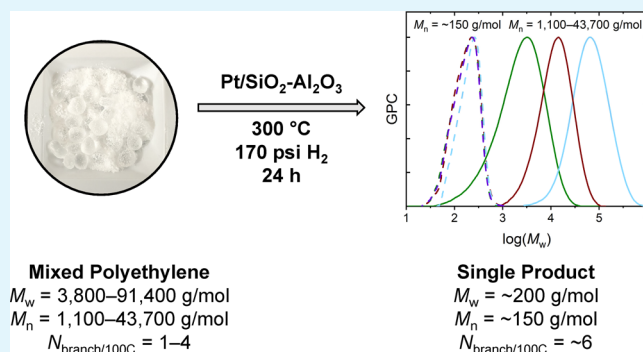
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ABSTRACT: Supported platinum nanoparticle catalysts are known to convert polyolefins to high-quality liquid hydrocarbons using hydrogen under relatively mild conditions. To date, few studies using platinum grafted onto various metal oxide (M_xO_y) supports have been undertaken to understand the role of the acidity of the oxide support in the carbon–carbon bond cleavage of polyethylene under consistent catalytic conditions. Specifically, two Pt/ M_xO_y catalysts (M_xO_y = SrTiO₃ and SiO₂–Al₂O₃; Al = 3.0 wt %, target Pt loading 2 wt % Pt ~1.5 nm), under identical catalytic polyethylene hydrogenolysis conditions (T = 300 °C, $P(H_2)$ = 170 psi, t = 24 h; M_w = ~3,800 g/mol, M_n = ~1,100 g/mol, $\bar{D}$ = 3.45, $N_{branch/100C}$ = 1.0), yielded a narrow distribution of hydrocarbons with molecular weights in the range of lubricants (M_w = < 600 g/mol; M_n < 400 g/mol; $\bar{D}$ = 1.5). While Pt/SrTiO₃ formed saturated hydrocarbons with negligible branching, Pt/SiO₂–Al₂O₃ formed partially unsaturated hydrocarbons (<1 mol % alkenes and ~4 mol % alkyl aromatics) with increased branch density ($N_{branch/100C}$ = 5.5). Further investigations suggest evidence for a competitive hydrocracking mechanism occurring alongside hydrogenolysis, stemming from the increased acidity of Pt/SiO₂–Al₂O₃ compared to Pt/SrTiO₃. Additionally, the products of these polymer deconstruction reactions were found to be independent of the polyethylene feedstock, allowing the potential to upcycle polyethylenes with various properties into a value-added product.

KEYWORDS: platinum, hydrogenolysis, upcycling, polyolefins, silica–alumina



1. INTRODUCTION

Over 400 million tons of plastics are generated worldwide each year, with projections estimating that by 2050, this production rate will exceed 1 billion tons.^{1,2} Approximately 50% of these plastics are used in single-use disposable applications, with about 75% of the total annual plastic production discarded yearly, leading to widespread environmental concern. Of the discarded plastics, approximately 50% are sent to landfill, 20% are incinerated for energy recovery, generating greenhouse gases and toxic byproducts, 20% are mismanaged and lost to the environment, and only about 10% are recycled.² Polyolefins, particularly high- and low-density polyethylenes (HDPE and LDPE), represent the most significant fraction of all plastic waste (~30%).³ The strong sp^3 carbon–carbon bonds in polyolefins make them highly resistant to degradation; consequently, recycling these materials typically involves mechanically shredding them into smaller fragments, followed by blending with virgin polymers.⁴ Owing to the exposure to high shear forces and heat during this process, the

resulting polymers typically show diminished mechanical properties and are consequently downcycled into lower quality, lower-value materials that are challenging to recycle at the end of their life. Therefore, waste plastics are a largely untapped resource for producing chemicals and new materials. Polymer upcycling efforts aim to transform plastic waste into value-added products, such as fuels,^{5–7} lubricants,^{7–10} surfactants,¹¹ and new polymers^{12–14} by utilizing various approaches, including functionalization (e.g., oxidation^{13–15}), dehydrogenation,^{12,16,17} metathesis,^{5,12} and deconstruction.^{7,9,18–20} These materials can then be further processed into new products and used in a second life cycle.

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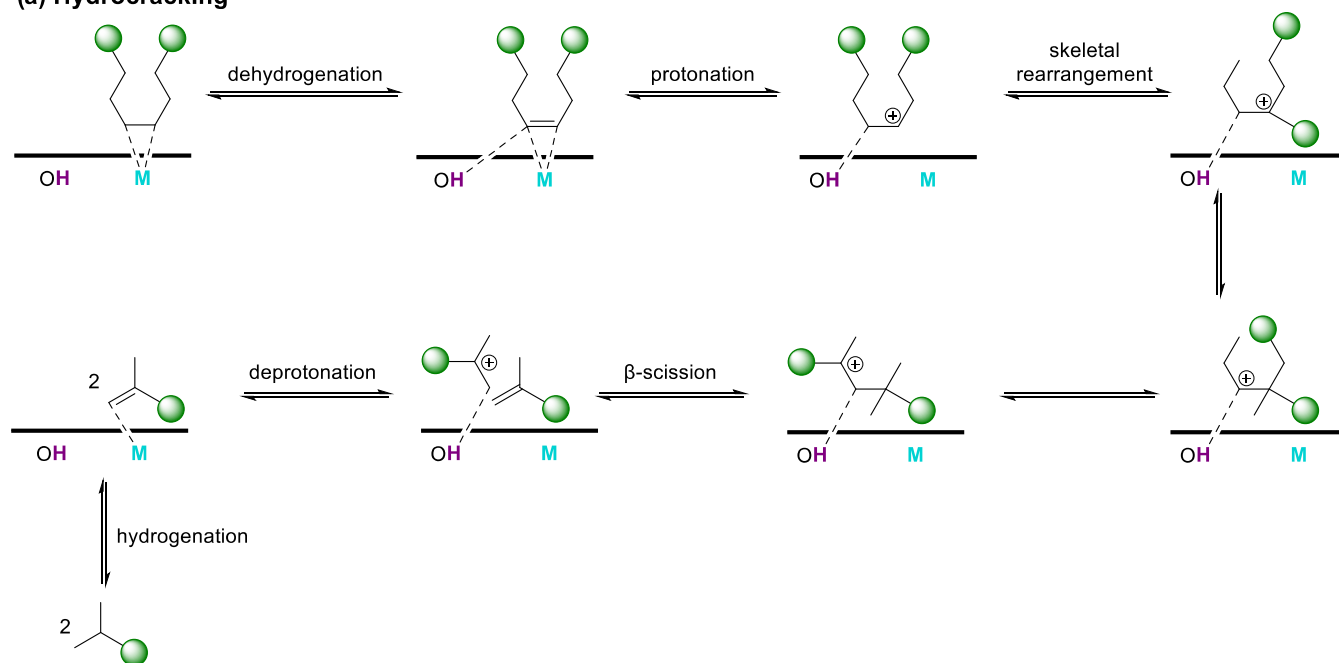
Table 1. Polyethylene Hydrogenolysis Catalysts and Conditions with the Corresponding Major Products

catalyst	polyethylene molecular weight		hydrogenolysis conditions				major products	
	M_w (g/mol)	M_n (g/mol)	Pt (wt %) ^a	temp (°C)	H ₂ pressure (psi)	time (h)	M_n (g/mol)	reference
Pt/SrTiO ₃	22,150	8150	0.27	300	170	18	2,050	18
Pt/SrTiO ₃	104,900	38,850	10	300	170	72	440	20
Pt/Al ₂ O ₃	22,150	8,150	0.10	300	170	18	1,850	18
Pt/Al ₂ O ₃	3,500	1,800	2.54	280	0	25	465 ^b	19
Pt/CeO ₂	4,000	1,700	0.15	260	870	24	none	26
Pt/SiO ₂	26,550	5,900	0.0013	250	130	24	250–360	7
mSiO ₂ /Pt/SiO ₂	26,550	5,900	0.0013	250	200	24	170–220	7
mSiO ₂ /Pt/SiO ₂	90,000	20,000	0.003	300	129	20	110–700	28
Pt/WZr	76,000		0.10	250	435	24	40–420 ^c	31
Pt/C	4,000	1,300	5	250	435	18	≤70 ^d	23
Pt/H-USY	4,000	1,700	0.03	260	870	42	70–290	26

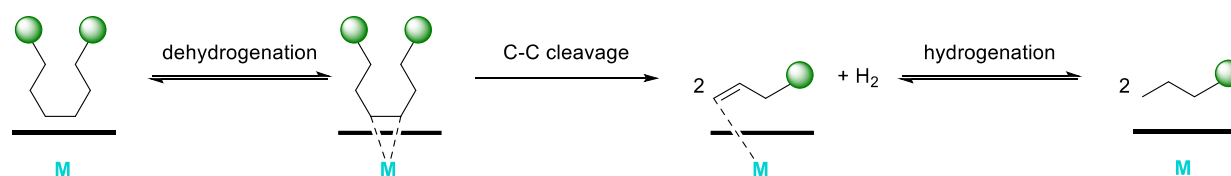
^aPt wt % with respect to PE. ^bContains alkyl aromatics. ^cProceeds via hydrocracking. ^dMinimal conversion.

Scheme 1. Representative (a) Hydrocracking and (b) Hydrogenolysis Mechanisms^a

(a) Hydrocracking



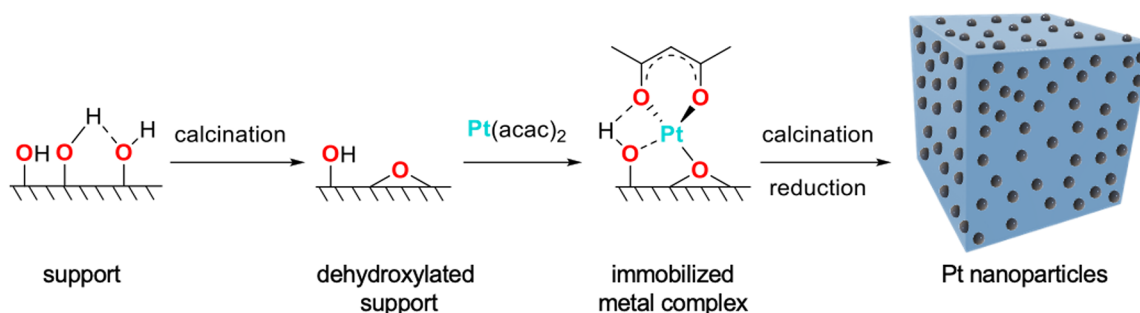
(b) Hydrogenolysis



^a[OH] corresponds to Brønsted acid sites and [M] to metal sites.^{22,36}

Existing deconstruction approaches, such as catalytic and thermal pyrolysis,⁴ effectively convert waste polymers into smaller fragments. However, these approaches typically require high operating temperatures (>500 °C), produce broad distributions of hydrocarbons (primarily C₁–C₄₀ olefins, cycloalkanes, dienes, and paraffins), and suffer from coke and char accumulation.⁴ Therefore, catalytic hydrogenolysis is an attractive polymer upcycling method to produce low molecular

weight, liquid hydrocarbons with narrow molecular weight distributions under relatively mild conditions.^{18,21} During this process, the C–C bonds along the polymer chain are cleaved in the presence of H₂, forming shorter alkane products. Hydrogenolysis catalysts are typically monofunctional species composed of metal nanoparticles deposited on metal oxide supports. The polyolefin and H₂ are absorbed onto the active

Scheme 2. Schematic of Proposed Catalyst Preparation³⁷

sites, facilitating C–C and H–H bond cleavage and C–H bond formation.²²

A range of metal/support combinations has been successfully implemented as hydrogenolysis catalysts, primarily focusing on ruthenium and platinum-based systems. Such catalysts include Ru/C for polyethylene,^{21,23,24} polypropylene,^{23,25} *n*-octadecane,²¹ and *n*-heptadecane²⁴ hydrogenolysis, Ru/CeO₂ for polyethylene hydrogenolysis,²⁶ Ru/TiO₂ for polypropylene hydrogenolysis,⁹ and Ru/Nb₂O₅ and Ru/ZrO₂ for the hydrogenolysis of aromatic polymers.²⁷ For platinum-based hydrogenolysis catalysts, research has primarily focused on Pt/Al₂O₃,^{18,19} Pt/C,²³ and Pt/SiO₂.^{7,28} based systems, with inconsistent hydrogenolysis conditions making direct comparison between different Pt/support combinations challenging (Table 1).²⁹ In a pioneering work,^{18,20,30} Pt nanoparticles (NPs) supported on strontium titanate (SrTiO₃) perovskite nanocuboids by atomic layer deposition (ALD) were shown to be highly active hydrogenolysis catalysts, converting polyethylene (PE; *M*_w = 17,200–420,000 g/mol and *M*_n = 8,000–158,000 g/mol) into high-quality liquid hydrocarbon products (*M*_w = 500–960 g/mol and *M*_n = 480–820 g/mol) under relatively mild conditions with 46–99% yield (300 °C, 170 psi H₂, solvent-free).^{8,18,20} In the same system, postconsumer plastic waste, including single-use plastic bags (*M*_w = 115,150 g/mol and *M*_n = 33,000 g/mol)¹⁸ and bubble wrap (*M*_w = 824,200 g/mol),⁸ was converted into value-added lubricants and waxes (*M*_w = 450–1,130 g/mol and *M*_n = 460–990 g/mol) with properties similar to synthetic base oils, such as polyalphaolefins (PAOs), and enhanced properties compared to petroleum-based lubricants, such as Group III mineral oil.⁸ Additionally, life cycle and techno-economic analyses found this hydrogenolysis process to be energetically efficient and potentially economically feasible.⁸

A significant factor in determining whether polymer deconstruction in the presence of H₂ proceeds via hydrogenolysis or hydrocracking mechanisms lies in the properties of the catalyst, where hydrocracking catalysts are typically bifunctional species composed of metal and acid sites.^{32–34} Polyolefin hydrocracking typically occurs via two or three consecutive steps (dehydrogenation of the polyolefin followed by C–C bond cracking and simultaneous or successive hydrogenation), where the acidic support is responsible for cracking and isomerization, and the metal sites are responsible for dehydrogenation/hydrogenation (Scheme 1a).³⁵ If significant degrees of unsaturation are present in the product, then it is likely that the hydrogenation step did not occur and polymer deconstruction proceeds via a monofunctional hydrocracking route.³² In contrast, hydrogenolysis catalysts are typically monofunctional species composed of metal precursors

dispersed over support materials, with bond breaking and hydrogenation occurring at the same site (Scheme 1b).²²

Despite several reports of supported platinum species as hydrogenolysis catalysts, there has yet to be a comprehensive study using platinum nanoparticles dispersed on metal oxide (M_xO_y) supports under consistent polyethylene deconstruction conditions. Herein, we report C–C bond cleavage of polyethylene using six Pt/M_xO_y catalysts under identical conditions to evaluate the role of the acidity of the supports on polymer deconstruction. Catalyst activity and polymer properties are assessed, finding that platinum dispersed on silica–alumina (SiO₂–Al₂O₃) and strontium titanate (SrTiO₃) give the highest yields of liquid/wax products (>60%) with monomodal molecular weight distributions (*M*_n < 400 g/mol). Additionally, further investigations into the products of these reactions suggest that polyolefin deconstruction proceeds predominantly via hydrogenolysis for Pt/SrTiO₃ and hydrocracking for Pt/SiO₂–Al₂O₃.

2. RESULTS AND DISCUSSION

2.1. Catalyst Preparation and Characterization. The Pt/M_xO_y catalysts were synthesized using a scalable chemical grafting procedure,^{37,38} which is complementary to atomic layer deposition (ALD)^{8,18,20} and strong electrostatic adsorption (SEA)³⁹ methods previously reported for Pt/SrTiO₃ synthesis. As the platinum precursor initially interacts with the hydroxyl groups on the support surface (Scheme 2), chemical grafting can allow for higher degrees of control than ALD or SEA methods, as the density of the metal sites can be somewhat controlled by varying the density of the hydroxide sites on the support surface (for example, by calcination).⁴⁰

Commonly used metal oxides, with a range of acidities and surface areas (Table 2), were employed as the support for

Table 2. Pt/M_xO_y Catalysts Studied for Hydrogenolysis and Their Corresponding Characterizing Data

catalyst	Pt ^a (wt %)	metallic surface area ^b (m ² /g)	Lewis acid sites ^c (μmol/g)	Brønsted acid sites ^c (μmol/g)
Pt/SrTiO ₃	0.49	240	-	0
Pt/SiO ₂	1.35	6	0	0
Pt/Al ₂ O ₃	1.40	9	94 ⁴⁵	0 ⁴⁵
Pt/SiO ₂ –Al ₂ O ₃	1.72	120	91	27
Pt/CeO ₂	1.24	83	24	2
Pt/TiO ₂	1.31	19	-	-

^aDetermined by inductively coupled plasma optical emission spectrometry (ICP-OES). ^bDetermined by CO pulse chemisorption titration. ^cDetermined by monitoring adsorbed pyridine on the catalyst surface by IR spectroscopy.

platinum nanoparticles. This methodology was used to compare eight different commercially available supports and study the role of their surfaces with the goal of differentiating possible operative mechanisms of C–C bond cleavage of polyolefins (e.g., hydrogenolysis vs hydrocracking) under the same grafting conditions, thus minimizing variables. However, the surface properties (e.g., acidity) of these oxides gave rise to a large range of platinum nanoparticle sizes and density, making the overall catalytic comparison challenging. It is noteworthy that only two supports (SrTiO₃ and SiO₂–Al₂O₃) yielded comparable Pt nanoparticles, allowing detailed catalytic studies (vide infra). Future work will address other deposition methods in order to provide uniform particle size after extensive optimization of the support, platinum precursor, and grafting conditions.

For the chemical grafting step, Pt(acac)₂ (acac = acetylacetonate; target loading 2 wt % Pt; 1 cycle of Pt deposition for all catalysts except Pt/SrTiO₃, which underwent 2 deposition cycles) and M_xO_y (SrTiO₃, SiO₂, Al₂O₃, SiO₂–Al₂O₃, TiO₂, CeO₂, ZrO₂, and HfO₂) were heated at 80 °C for 3 days, followed by calcination in air at 300 °C for 4 h and reduction under 10% hydrogen at 300 °C for 4 h. The Pt/M_xO_y catalysts studied are listed in Table 2, where SiO₂–Al₂O₃ (aluminum-doped mesoporous MCM-41) contains 3.0 wt % Al (see the Supporting Information for a second example of a silica–alumina catalyst deposited on Davicat 3113 with 7.6 wt % Al). The synthesis of Pt/ZrO₂ and Pt/HfO₂ was also attempted, but minimal Pt content was observed by inductively coupled plasma optical emission spectroscopy (ICP-OES) (0.07 and 0.11 wt %, respectively; Table S2).

For all catalysts, except Pt/SrTiO₃, the platinum content was determined as 1.25–1.75 wt % by ICP-OES (Table 2). The platinum content of Pt/SrTiO₃ is significantly lower (0.49 wt %), potentially due to surface species remaining post–pretreatment interfering with Pt grafting.³⁷ Brunauer–Emmett–Teller (BET) surface area analysis shows a broad range of surface areas between 25 and 845 m²/g (Table S5). With the exception of Pt/SiO₂, all catalysts showed smaller surface areas than the original M_xO_y support, indicating that the Pt nanoparticles block some of the pores. As expected, powder X-ray diffraction (PXRD) showed amorphous structures for SiO₂–Al₂O₃ and Pt/SiO₂–Al₂O₃ (Figure S2).⁴¹

High-angle annular dark-field imaging (HAADF) scanning transmission electron microscopy (STEM), energy-dispersive X-ray (EDX) mapping, temperature-programmed reduction (H₂-TPR), and CO chemisorption were used to study the catalyst microstructure. All catalysts show well-distributed platinum across the support surface by STEM (Figures S3–S9). However, the catalysts showed wide variations in particle size (between 1.0 and 10.4 nm by TEM), with Pt/SrTiO₃, Pt/SiO₂–Al₂O₃, and Pt/CeO₂ showing the smallest and most comparable nanoparticle size (approximately 1.5 nm; Table S3). These differences highlight the effect of the surface properties, such as acidity and/or number of anchoring sites, on the synthesis of nanoparticles under identical deposition conditions. It is also noted that the large differences in nanoparticle size could introduce structure sensitivity effects to the catalysis reactions, which is most prevalent for <10 nm particles, with smaller particles typically demonstrating higher activity.^{42,43}

H₂ temperature-programmed reduction (H₂-TPR) revealed different Pt-support interactions among these catalysts (Figure S10). A H₂ reduction peak at 68 °C for Pt/SiO₂ indicates a

weak interaction between Pt and the SiO₂ support. For Pt/TiO₂ and Pt/Al₂O₃, the reduction peaks increase to 200 and 280 °C, respectively, indicative of strong metal–support interactions. Notably, Pt/SiO₂–Al₂O₃ and Pt/CeO₂ show similar Pt-support interaction behavior, evidenced by the H₂ reduction peaks at approximately 160 °C. However, it is worth noting that several factors can influence H₂-TPR profiles, including catalyst pretreatment conditions, metal content, and reduction of the support. CO chemisorption shows similar metallic surface areas for Pt/SiO₂–Al₂O₃ and Pt/CeO₂ (120 and 83 m²/g, respectively), corresponding to the area of active Pt on the support surface. Pt/SrTiO₃ shows the highest metallic surface area (240 m²/g), followed by Pt/SiO₂–Al₂O₃ and Pt/CeO₂, with Pt/TiO₂, Pt/Al₂O₃, and Pt/SiO₂ showing a negligible metallic surface area (<20 m²/g).

The Lewis and Brønsted acidities of Pt/SiO₂–Al₂O₃, Pt/CeO₂, Pt/SiO₂, and Pt/SrTiO₃ were evaluated by monitoring adsorbed pyridine on the catalyst surface by infrared (IR) spectroscopy (Table 2 and Figure S11).^{44,45} Temperature-programmed desorption (TPD) of ammonia from the surface was initially attempted; however, inconsistencies in the results highlighted that pyridine is a better substrate for quantification of the acids of these Pt/M_xO_y species. Pt/SiO₂ displayed no Lewis acid sites (LAS) or Brønsted acid sites (BAS). For Pt/SrTiO₃, the peak at 1600 cm^{−1} likely corresponds to pyridine adsorbed on the LAS of SrTiO₃.⁴⁶ However, the 1,450 cm^{−1} peak typically used for quantification is too weak to allow accurate determination of LAS. Therefore, the LAS density is assumed to be low, with no BAS. Pt/CeO₂ primarily displays LAS (LAS = 24 μmol/g and BAS = 2 μmol/g). In contrast, Pt/SiO₂–Al₂O₃ contains significant degrees of Lewis and Brønsted acidity (LAS = 91 μmol/g and BAS = 27 μmol/g, respectively).

2.2. Polyethylene Deconstruction. Hydrogenolysis aims to produce high-quality, low molecular weight liquid/wax products from polyethylenes (target M_n < 600 g/mol and dispersity *D* ≤ 1.5) using H₂ to cleave C–C bonds. The six Pt/M_xO_y catalysts discussed previously were studied for polyethylene deconstruction in a 100 mL Parr autoclave reactor with an amount of Pt/M_xO_y corresponding to 0.004 g of Pt (calculated from ICP-OES) and 3.00 g of polyethylene (PE; M_w = ~3800 g/mol, M_n = ~1100 g/mol, M_w/M_n = ~3.5, N_{branch}/100C = 1.0; Figures S13 and S14). Reactants were placed into a glass sleeve inside the reactor. The reactor was then heated to 300 °C, at which time 170 psi H₂ was added, and the reactions were run for 24 h with a stirring rate of 800 rpm. Control reactions were also run under the same conditions with no catalyst and with the M_xO_y supports. Upon completion, the reactor was cooled to approximately 50 °C, and the headspace was directly analyzed by gas chromatography with flame ionization detection (GC-FID) (detection range: C₁–C₈ alkanes and alkenes). The reactor was then cooled to room temperature, opened to the atmosphere, and the liquid/wax products were isolated between the walls of the steel reactor and glass sleeve; over the course of the reaction, the lower molecular weight products vaporized out of the glass sleeve and condensed into the steel reactor. The products remaining in the glass sleeve were also isolated, washed with dichloromethane to remove any low molecular weight species, and analyzed. These products are referred to as gas, liquid/wax, and solid, encompassing C₁–C₆, C₇–C₃₀, and C₃₀₊, respectively. The gaseous yield was calculated as a mass

Table 3. Products from the Catalytic Hydrogenolysis of Polyethylene^a

catalyst	gas		liquid/wax				solid yield (%)
	yield ^b (%)	yield ^c (%)	M_w^d (g/mol)	M_n^d (g/mol)	$\bar{D}^d$	$N_{branch/100C}^e$	
SrTiO ₃	14	0					86
Pt/SrTiO ₃	25	54	310	230	1.3	0.1	21
SiO ₂	9	3	320	220	1.5		88
Pt/SiO ₂	14	2	570	370	1.5	0.2	84
Al ₂ O ₃	2	1	4,350	2,610	1.7		97
Pt/Al ₂ O ₃	11	4	340 ^f	250 ^f	1.4 ^f	0.4	85
			6,730	4,330	1.6		
			800 ^f	620 ^f	1.3 ^f		
SiO ₂ –Al ₂ O ₃	34	30	290	250	1.2	5.0	36
Pt/SiO ₂ –Al ₂ O ₃	28	66	190	130	1.5	5.5	6
CeO ₂	12	3	3,310	2,190	1.5		85
Pt/CeO ₂	10	3	440 ^f	330 ^f	1.3 ^f	0.5	87
			340	270	1.3		
TiO ₂	12	1	350	270	1.3		87
Pt/TiO ₂	9	2	350	300	1.2	1.8	89
no catalyst	9	3	550	320	1.7	0.3	88

^aReaction conditions: PE (3.00 g; $M_w = \sim 3,800$ g/mol, $M_n = \sim 1,100$ g/mol, $M_w/M_n = \sim 3.5$, $N_{branch/100C} = 1.0$), Pt/ M_xO_y (equivalent to 4 mg of Pt), or equivalent quantity of M_xO_y , 300 °C, 24 h, and 170 psi H₂. ^bDetermined by mass balance: gas % yield = 100 – liquid/wax % yield – solid % yield. ^cDefined as the weight (g) of hydrocarbons recovered relative to the initial mass of PE. ^dDetermined by high-temperature GPC with RI detection. ^eDetermined by ¹³C{¹H} NMR spectroscopy. ^fBimodal molecular weight distribution, as determined by HT-GPC-RI.

balance considering the solid and liquid/wax yields and mass of the starting polyethylene (Table 3 and Figure S19).

2.2.1. Gas Products. Of the six catalysts, Pt/SiO₂–Al₂O₃ and Pt/SrTiO₃ give the highest overall conversion to products: 94 and 79%, respectively. Regarding the gaseous products, Pt/SiO₂–Al₂O₃ and Pt/SrTiO₃ showed the highest yield (28 and 25%, respectively), as expected, based on the higher overall conversion of these catalysts (Figure 1). All other catalysts gave

dominant polyolefin deconstruction pathway.³² Conversely, for Pt/SiO₂–Al₂O₃, the dominant gaseous products were propane (60% of the total gas yield) and isobutane (12% of the total gas yield), with the formation of these C₃ and C₄ alkanes suggesting hydrocracking as the major polyolefin deconstruction pathway (Figure 2).³²

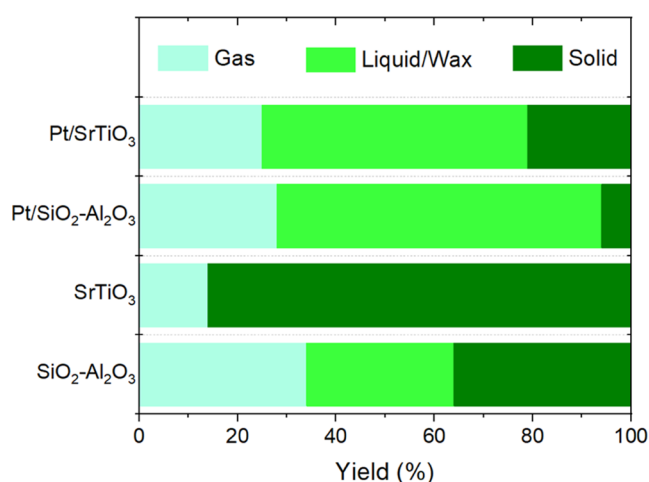


Figure 1. Gas, liquid/wax, and solid yield for the hydrogenolysis of PE ($M_w = \sim 3,800$ g/mol, $M_n = \sim 1,100$ g/mol) using Pt/SrTiO₃, Pt/SiO₂–Al₂O₃, SrTiO₃, and SiO₂–Al₂O₃. Hydrogenolysis conditions: PE (3.00 g), Pt/ M_xO_y (equivalent to 4 mg of Pt), or equivalent quantity of M_xO_y , 300 °C, 24 h, and 170 psi H₂.

gaseous product yields similar to control reactions (9–14%). The gas products were exclusively alkanes, primarily $\leq C_4$, resulting from near-chain end hydrogenolysis (Figure S21).^{47,48} The major gas product for all Pt-catalyzed conversions, except Pt/SiO₂–Al₂O₃, was methane (35–75% of the total gas yield), indicating that hydrogenolysis is the

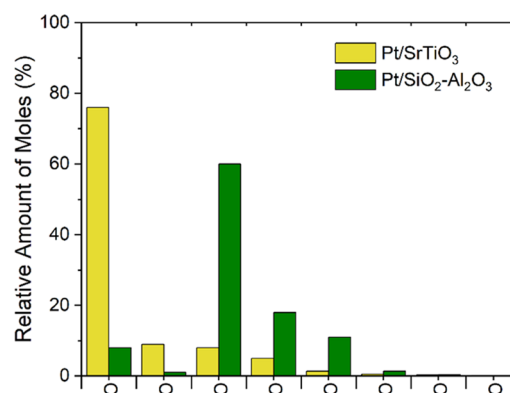


Figure 2. Representative distribution of C₁–C₈ alkanes produced during the hydrogenolysis of PE ($M_w = \sim 3800$ g/mol, $M_n = \sim 1100$ g/mol) using Pt/SrTiO₃ (yellow) and Pt/SiO₂–Al₂O₃ (green), as determined by GC-FID. Hydrogenolysis conditions: PE (3.00 g), Pt/ M_xO_y (equivalent to 4 mg of Pt), 300 °C, 24 h, and 170 psi H₂.

The gas yield of Pt/SrTiO₃ is higher than previously reported for ALD and SEA methods (<10%),^{8,18} potentially due to differences in the Pt nanoparticle formation owing to differences in the synthetic procedures.³⁷ In contrast to the Pt/ M_xO_y catalysts, the gas product composition of the control reaction without any catalyst contained 94% alkanes (primarily $\leq C_4$) and 6% alkenes (propylene and isobutylene; Figure S23) as a result of thermal cracking (i.e., pyrolysis).⁴⁹ The control reactions using M_xO_y also showed the presence of alkenes as the lack of metal sites hinders hydrogenation, with the high C₃

and C₄ production indicating a cracking mechanism (Figure S24).

2.2.2. Solid Products. The solid products remaining in the glass sleeve upon completion of the hydrogenolysis reactions were analyzed by high-temperature gel-permeation chromatography with refractive index detection (HT-GPC-RI). In general, these products show molecular weights consistent with the starting material, within error margins (Table S6 and Figures S30 and S31). However, GPC shows that the lower molecular weight tail is reduced, suggesting that the lower molecular weight species have been extracted during the workup. For catalysts with low conversion, representative ¹H and ¹³C{¹H} NMR spectra of the solid product remaining in the glass sleeve showed no change in the branch density or identity compared to the starting material, confirming primarily unreacted starting material (demonstrated for Pt/SiO₂: 16% conversion under standard condition; Figures S32 and S33). For catalysts with higher conversions, the remaining solid was primarily composed of unreacted starting material, with the increase in branch density from $N_{\text{branch}/100\text{C}} = 1$ to ≈ 4 highlighting the formation of C₁ branches (demonstrated for Pt/SiO₂-Al₂O₃ with 26% conversion after 6 h and 66% conversion after 18 h; Figures S34-S37).

2.2.3. Liquid/Wax Products. Pt/SiO₂-Al₂O₃ and Pt/SrTiO₃ give the highest yield of liquid/wax products (66 and 64%, respectively). However, Pt/SiO₂-Al₂O₃ increases product branching while Pt/SrTiO₃ decreases product branching ($N_{\text{branch}/100\text{C}} = 5.5$ and 0.1, respectively, compared to $N_{\text{branch}/100\text{C}} = 1.0$ for the starting PE). The other platinum catalysts, Pt/SiO₂, Pt/Al₂O₃, Pt/TiO₂, and Pt/CeO₂, show minimal conversion to liquid/wax products, giving similar yields to the control reactions (<5%; Table 3 and Figure 1). To better understand the vast differences in polyethylene deconstruction ability, the properties of the Pt/M_xO_y catalysts need to be considered.

Hydrogenolysis is a structure-sensitive reaction where catalyst particle size significantly affects activity.¹⁸ For example, previous work using Pt/SrTiO₃ for polyethylene hydrogenolysis found that smaller Pt nanoparticles with higher concentrations of undercoordinated Pt sites showed increased polyethylene hydrogenolysis activity, leading to overhydrogenolysis of the polyethylene to light hydrocarbons.¹⁸ In this work, Pt/SrTiO₃ with average nanoparticle sizes of 1.2 nm completely transformed PE into gaseous hydrocarbons (C₁-C₈), while Pt/SrTiO₃ with average nanoparticle sizes of 2.3 and 2.9 nm, respectively, converted PE into liquids product ($M_n \approx 675$ g/mol). Recently, Wu et al. also showed that a catalytic architecture composed of a mesoporous silica shell surrounding platinum nanoparticles supported on a solid silica sphere (mSiO₂/Pt/SiO₂) showed similar trends.²⁸ In this system, polyethylene hydrogenolysis activity increased with decreasing size of the platinum nanoparticles, with activity following the trend: mSiO₂/Pt-1.7 nm/SiO₂ > mSiO₂/Pt-2.9 nm/SiO₂ > mSiO₂/Pt-5.0 nm/SiO₂. In this work, the average nanoparticle diameters of Pt/SiO₂-Al₂O₃, Pt/SrTiO₃, and Pt/CeO₂ are similar (<2 nm, Table S3) and are, therefore, the most comparable. However, these catalysts display vast differences in activity, indicating that nanoparticle size may not be the primary factor influencing activity. Additionally, there appears to be no correlation between the BET surface area and catalyst activity; surface areas of 20 and 845 m²/g for the two most active catalysts (Pt/SrTiO₃ and Pt/SiO₂-Al₂O₃). However, the catalysts with the highest active surface area (Pt/

SrTiO₃ and Pt/SiO₂-Al₂O₃ with a metallic surface area of 0.95 and 0.48 m², respectively, in the hydrogenolysis reaction) show the highest activity (79 and 94% total conversion), while Pt/CeO₂ has a similar active metallic surface area (0.33 m² in the hydrogenolysis reaction), but negligible activity (13%, comparable to no catalyst). In contrast, Pt/TiO₂, Pt/Al₂O₃, and Pt/SiO₂ have large (>5 nm) Pt particles with a negligible metallic surface area (<0.07 m²) and show low activity (<16% total conversion).

Pt/SrTiO₃ and Pt/CeO₂ (Figure S19) are the two most similar catalysts in terms of nanoparticle diameter (≈ 1 nm), active metallic surface area (0.95 and 0.33 m²), and Brønsted acidity (negligible). However, their ability to deconstruct polyethylene varies greatly (79 and 13% total conversion, respectively). Despite their similar properties, Pt/SrTiO₃ is able to more efficiently deconstruct polyethylene via hydrogenolysis. The high activity of Pt/SrTiO₃ has previously been ascribed to the close lattice-match between the SrTiO₃ and Pt, resulting in a cube-on-cube epitaxy that produces a large interfacial energy term and leads to enhanced Pt nanoparticle stability,¹⁸ while Nakaji et al. have also observed low conversion for Pt/CeO₂ under similar conditions (<0.1% conversion of LDPE, 260 °C, 145 psi H₂, 24 h).²⁶ Furthermore, the high activity of Pt/SrTiO₃ synthesized by chemical grafting is comparable to Pt/SrTiO₃ synthesized by ALD and SEA, which were tested at higher Pt loadings and reaction times; 54, 42-99, and 99% conversion to liquid/wax products for Pt/SrTiO₃ synthesized via chemical grafting, ALD and SEA, respectively.^{18,39} This highlights the effects of both reaction conditions and deposition method on catalyst activity.

Catalyst acidity gives further insight into activity trends and the polyethylene deconstruction pathways. As noted previously, Lewis acidity follows the order Pt/SiO₂ < Pt/SrTiO₃ < Pt/CeO₂ < Pt/Al₂O₃ $\approx$ Pt/SiO₂-Al₂O₃ and Brønsted acidity follows the order Pt/SiO₂ $\approx$ Pt/SrTiO₃ $\approx$ Pt/Al₂O₃ $\approx$ Pt/CeO₂ $\ll$ Pt/SiO₂-Al₂O₃ with Pt/SiO₂ showing no Lewis or Brønsted acidity (note that Pt/Al₂O₃ acidity is assumed to be similar to Sun et al. for Pt/Al₂O₃ synthesized via wetness impregnation with a Pt loading of 1.5 wt %: LAS of 94 $\mu\text{mol/g}$, and BAS of 0 $\mu\text{mol/g}$).⁴⁵ Considering the low conversions of Lewis acidic of Pt/CeO₂ and Pt/Al₂O₃, Brønsted acidity is determined to have a more significant effect on catalyst activity. The lack of Brønsted acidity in Pt/SiO₂, Pt/SrTiO₃, Pt/Al₂O₃, and Pt/CeO₂ suggests that polymer deconstruction can only occur through a hydrogenolysis pathway. In contrast, the higher Brønsted acidity of Pt/SiO₂-Al₂O₃ suggests that polymer deconstruction likely follows a hydrocracking pathway as the primary deconstruction mechanism.^{50,51}

HT-GPC-RI analysis of the liquid/wax products produced using Pt/M_xO_y and M_xO_y showed monomodal product formation for all catalysts except Pt/Al₂O₃-Al₂O₃ and CeO₂ (Figures 3 and S78). The bimodal molecular weight distribution for Pt/Al₂O₃-Al₂O₃ and CeO₂ indicates possible thermal decomposition of the starting PE; however, this was not investigated further due to the low conversion to liquid/wax products (<15%). In general, the molecular weights of the hydrogenolysis products decreased from $M_w = \sim 3,800$ to <600 g/mol and from $M_n = \sim 1100$ to <400 g/mol, with a narrowing of dispersity from $\bar{D} = 3.6$ to <1.6. Owing to the higher activity of Pt/SiO₂-Al₂O₃ compared to the other catalysts, further analysis of the liquid product was conducted. As the low molecular weights of the hydrogenolyzed products were at the limit of the HT-GPC-RI calibration range, the molecular

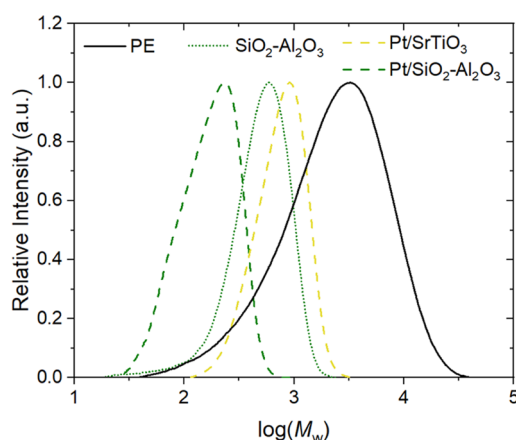


Figure 3. Molecular weight distribution plots for starting PE (solid black; $M_w = 3,800$ g/mol, $M_n = 1100$ g/mol) and hydrogenolyzed PE using Pt/SrTiO₃ (yellow dash), Pt/SiO₂-Al₂O₃ (green dash), and SiO₂-Al₂O₃ (green dot). Hydrogenolysis conditions: PE (3.00 g), Pt/ M_xO_y (equivalent to 4 mg of Pt), or equivalent quantity of M_xO_y , 300 °C, 24 h, and 170 psi H₂.

weights were more accurately determined using ultraperformance liquid chromatography (UPLC), which showed $M_w = 320$ g/mol, $M_n = 290$ g/mol, and $D = 1.1$.

For all catalysts, except Pt/SiO₂-Al₂O₃ and Pt/TiO₂, NMR spectroscopy analysis of the liquid/wax products shows decreased branch density compared to the PE starting material (Table 3 and Figures S38–S52). Consistent with the starting PE, only C₄ branches were observed in the products, suggesting near-chain-end hydrogenolysis on the branches or C–C bond scission at the branch points along the chain. The liquid/wax products formed using Pt/SiO₂-Al₂O₃ show increased branch density compared to the starting material (from $N_{\text{branch}/100\text{C}} = 1.0\text{--}5.5$) and increases in the type of branching (C₁–C₄ and *sec*-butyl branches; Figure S48), which suggests that hydrocracking is occurring alongside hydrogenolysis. In contrast to hydrogenolysis, where monofunctional catalysts enable C–C cleavage and C–H bond formation to occur at the same site, hydrocracking typically employs bifunctional catalysts where the acidic support is responsible for C–C bond scission, and the metal sites hydrogenate the olefinic intermediates. This can allow skeletal isomerization to occur at the acid sites, leading to an increase in branching.²²

The occurrence of a hydrocracking mechanism can be further confirmed by the presence of small amounts of terminal and internal olefins in the ¹H NMR spectrum of the Pt/SiO₂-Al₂O₃ liquid/wax products (vide infra). Furthermore, the ¹H NMR spectrum of the liquid products obtained using Pt/SiO₂-Al₂O₃ shows the formation of small amounts of aromatics as a result of olefin cyclization reactions (aromatic protons at 6.5–9.0 ppm and CH_α protons bound to aromatic rings at 2.0–3.5 ppm).⁵² Further analysis shows that alkylbenzenes are the major products (6.5–7.4 ppm), with fewer fused aromatic ring products such as alkylnaphthalenes and alkylphenanthrenes (7.4–9.0 ppm).^{19,45} Species with –CH₂R substituents (i.e., Ph–CH₂R) were found to be the major products, as more intense signals are observed in the methylene (CH₂) region (2.5–3.1 ppm) compared to the methine (CH) and methyl (CH₃) regions. The ratio of protons associated with an aliphatic carbon directly bound to an aromatic ring (Ph–CH_α) relative to the aromatic protons (H_{aromatic}) was calculated as 1.5 for Pt/SiO₂-Al₂O₃. The same

ratio can be calculated for a range of potential products (Figure S77), indicating that the major species are, on average, trialkylbenzenes.⁴⁵ The aromatic selectivity can also be determined, showing a selectivity of 96 mol % alkanes and 4 mol % aromatics (2 mol % alkylbenzenes +1 mol % alkylnaphthalenes +1 mol % alkylphenanthrenes) for Pt/SiO₂-Al₂O₃. Analysis by room temperature GPC with refractive index detection (RT-GPC-RI) showed molecular weights similar to room temperature GPC with ultraviolet detection (RT-GPC-UV), $M_w = 230$ g/mol and $M_n = 210$ g/mol, and $M_w = 230$ g/mol and $M_n = 200$ g/mol, respectively, indicating that the UV-active aromatic chromophores are evenly distributed across the molecular weight range (Table S7).¹⁹

Overall, Pt/SiO₂-Al₂O₃ and Pt/SrTiO₃ show the highest conversion of polyethylene to gas and liquid/wax products. This is likely due to their small nanoparticle diameter (<1 nm), with the higher Brønsted acidity of Pt/SiO₂-Al₂O₃ enabling polymer deconstruction via hydrocracking and the higher stability of Pt/SrTiO₃ enabling deconstruction via hydrogenolysis. The overall rate of carbon–carbon bond scission ($r_{\text{C-C}}$), which is independent of the nature of the active site (e.g., Pt and H⁺), for these two catalysts was determined using the method described by Sun et al. (see vide infra for methods and limitations).⁴⁵ The $r_{\text{C-C}}$ was approximately 50% faster for Pt/SrTiO₃ than Pt/SiO₂-Al₂O₃ (2.1 and 1.4 mmol/h, respectively) due to increased methane production. The turnover frequency (TOF) for Pt/SiO₂-Al₂O₃ cannot be calculated because Pt is not solely responsible for the reaction rate (acid site catalyzed hydrocracking). On the other hand, because Pt is the only catalytic hydrogenolysis site on SrTiO₃, the TOF is 0.29 s^{−1} (Supporting Information and Table S4 for details). This TOF value is comparable to hexane hydrocracking using Pt/Al₂O₃ (TOF = 0.30 s^{−1} at 300 °C).^{53,54} In contrast, ethane and *n*-hexane hydrogenolysis using Pt clusters on SiO₂ is significantly slower than the values reported here (TOF of approximately 10^{−4} s^{−1} at 320 °C).⁴⁸

2.2.4. Recyclability of Pt/SiO₂-Al₂O₃. A key advantage of Pt/SrTiO₃ over other Pt/ M_xO_y systems is the relatively small changes in its structure post catalysis, enabling it to be reused without significant changes to the product composition. Recently, McCullough et al. studied the recyclability of Pt/SrTiO₃ synthesized via chemical grafting for isotactic polypropylene (iPP) hydrogenolysis. 80–90% conversion to liquid/wax products was achieved for 4 recycling cycles, with only slight variations in product molecular weights.³⁷ Analysis of the as-prepared and recycled catalyst showed consistent particle size (1.0–1.3 nm), Pt loading (0.936–0.960 wt %), and Pt/Ti ratio (0.0214–0.0374). In contrast, reusing Pt/SiO₂-Al₂O₃ after one cycle of hydrogenolysis led to decreased overall conversion (from 94 to 75%), increased gas yield (from 28 to 43%), and significantly reduced liquid/wax yield (from 66 to 22%), which is likely a result of changes in particle size and catalyst acidity. STEM images of the fresh and recycled catalyst show that the Pt is very well distributed in small nanoparticles before catalysis. However, post reaction, the Pt is sintered across the support with a mixture of small and large nanoparticles (an increase in average nanoparticle size from 1.9 to 8.5 nm), which is consistent with the observed decrease in conversion (Figure 4). This behavior is not observed for Pt/SrTiO₃, in which the cube-on-cube epitaxy of the platinum on the support prevents the sintering of the nanoparticles.⁵⁵

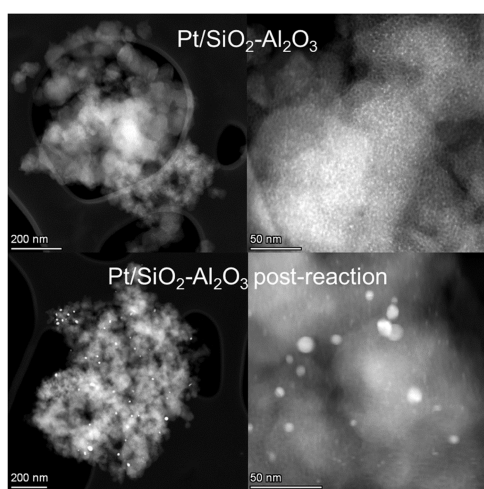


Figure 4. HAADF STEM images of the as-prepared Pt/SiO₂–Al₂O₃ (top) and Pt/SiO₂–Al₂O₃ post reaction (bottom).

The increased amount of methane (from 8 to 48%) and reduced amount of propane (60 to 17%) produced as a fraction of the overall gas product for recycled Pt/SiO₂–Al₂O₃ compared to the fresh catalysts suggests changes in the acid sites during the first catalytic cycle, leading to an increase in hydrogenolysis relative to hydrocracking (Figure S25). Monitoring pyridine after desorption from the catalyst surface by IR confirms a loss of acidity for Pt/SiO₂–Al₂O₃ post reaction (LAS decreases from 91 to 68 μmol/g and BAS decreases from 27 to 11 μmol/g; Figure S12). Note that the determination of the BAS density may carry some error as the BAS resonance at 1545 cm^{−1} used for quantification overlaps with an intense broad peak stemming from hydrocarbons remaining on the catalyst surface. Additionally, strong signals in the 2,900–3,100 cm^{−1} range corresponding to C–H stretching vibrations indicate the presence of hydrocarbons on the surface, which can potentially block active sites and decrease the activity of the recycled catalyst. In addition, the liquid product of the recycled catalyst shows a bimodal molecular weight distribution, potentially suggesting the presence of multiple types of active sites; *M_w* increased from 190 g/mol for the fresh catalyst to 7110 g/mol for the lowest molecular weight peak of the recycled catalyst, with corresponding *M_n* values of 130 g/mol and 640 g/mol, respectively (Figure S80).

2.2.5. Reaction Time Effects Using Pt/SiO₂–Al₂O₃. The effect of reduced reaction time on polyethylene deconstruction was studied using Pt/SiO₂–Al₂O₃ under consistent hydrogenolysis conditions (*T* = 300 °C, *P*(H₂) = 170 psi, *t* = 6–24 h; Figure 5). Interestingly, the yield of gaseous products is relatively unaffected by reaction time (23, 26, and 28% yield for 6, 18, and 24 h, respectively), indicating that overhydrogenolysis does not occur for these systems due to the isolation of the products away from the catalyst. However, the distribution of gaseous products changes, with a shift to smaller hydrocarbon species at longer reaction times (Figure S22).

As expected, the overall conversion and yield of liquid/wax products decrease with reduced reaction time. However, the molecular weights of the liquid/wax products are unchanged (Figure S79). The rate of carbon–carbon bond scission (*r_{C–C}*) was determined, which was found to decrease as the reaction time increased (3.0, 1.5, and 1.4 mmol/h after 6, 12, and 18 h,

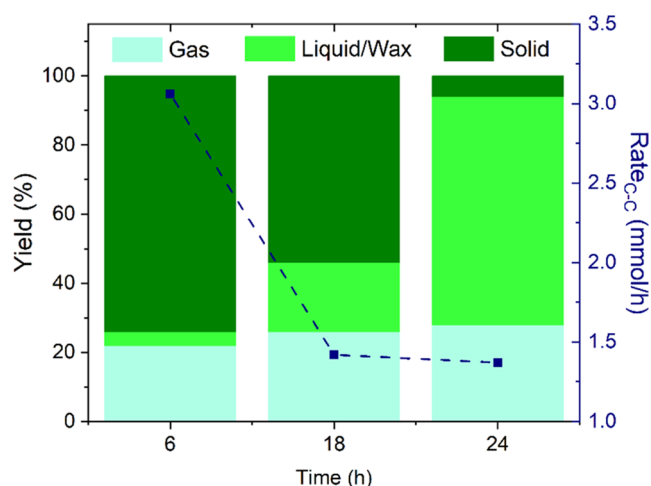


Figure 5. Gas, liquid/wax, and solid yield for the hydrogenolysis of PE (*M_w* = ~3,800 g/mol, *M_n* = ~1,100 g/mol) using Pt/SiO₂–Al₂O₃. The rate of C–C bond scission is shown in navy. Hydrogenolysis conditions: PE (3.00 g), Pt/*M_xO_y* (equivalent to 4 mg of Pt), 300 °C, and 170 psi H₂.

respectively).⁴⁵ This may be due to the decreased H₂ content at longer reaction times, slowing the reactivity.

2.2.6. Background Reactivity of SiO₂–Al₂O₃. To better understand the background reactivity of the support, polyethylene deconstruction was conducted using SiO₂–Al₂O₃ as the catalyst. Under consistent hydrogenolysis conditions (*T* = 300 °C, *P*(H₂) = 170 psi, *t* = 24 h), conversion similar to gaseous products (approximately 30%) and reduced conversion to liquid products were observed compared to Pt/SiO₂–Al₂O₃ (66 and 30%, respectively), highlighting the necessity of the Pt active sites for the production of products with molecular weights in the range of lubricants (Table 2 and Figure 1). Similarly, *r_{C–C}* was slightly higher for Pt/SiO₂–Al₂O₃ than SiO₂–Al₂O₃ (1.4 and 1.2 mmol/h, respectively), further highlighting the background reactivity of SiO₂–Al₂O₃. A similar effect has been observed for Pd grafted onto a Hβ zeolite for polyethylene pyrolysis, where conversion to liquid products increased from 48 to 98% yield for the Pd-impregnated catalyst compared to the Hβ support.⁵⁶ Liu et al. also noted an increase in the activity of Pt/WO₃/ZrO₂ mechanically blended with HY(30) zeolite compared to HY(30) alone for the cracking of LDPE in the presence of hydrogen, with 7 and 91% unreacted solid residue remaining after 2 h, respectively.⁶ In addition, Sun et al. reported an increase in the activity of Pt grafted onto Hβ zeolite compared to the Hβ zeolite alone for the hydrogenolysis of LDPE into C₅–C₁₂ gasoline range alkanes, showing 55 and 16 wt % conversions, respectively.⁵⁷ The authors propose that this is due to Pt hydrogenating the intermediates and suppressing the formation of char residues due to the facilitation of C–C bond breaking under a H₂ atmosphere, likely also operative in this system. Furthermore, Tan et al. also show that polyethylene depolymerization on metal-free mordenite framework inverted (MFI) zeolites proceeds at higher rates than the analogous Pt or Ni-loaded MFI zeolites.⁵⁸

The liquid/wax products produced using SiO₂–Al₂O₃ show features similar to the products obtained using Pt/SiO₂–Al₂O₃; *M_w* = 230 g/mol, *M_n* = 160 g/mol, *N_{branch}/100C* = 5.0 (C₁–C₄ and *sec*-butyl branches), and Ph–CH₂/H_{aromatic} = 1.8 (Figures S53–S55). In particular, the similar branch density

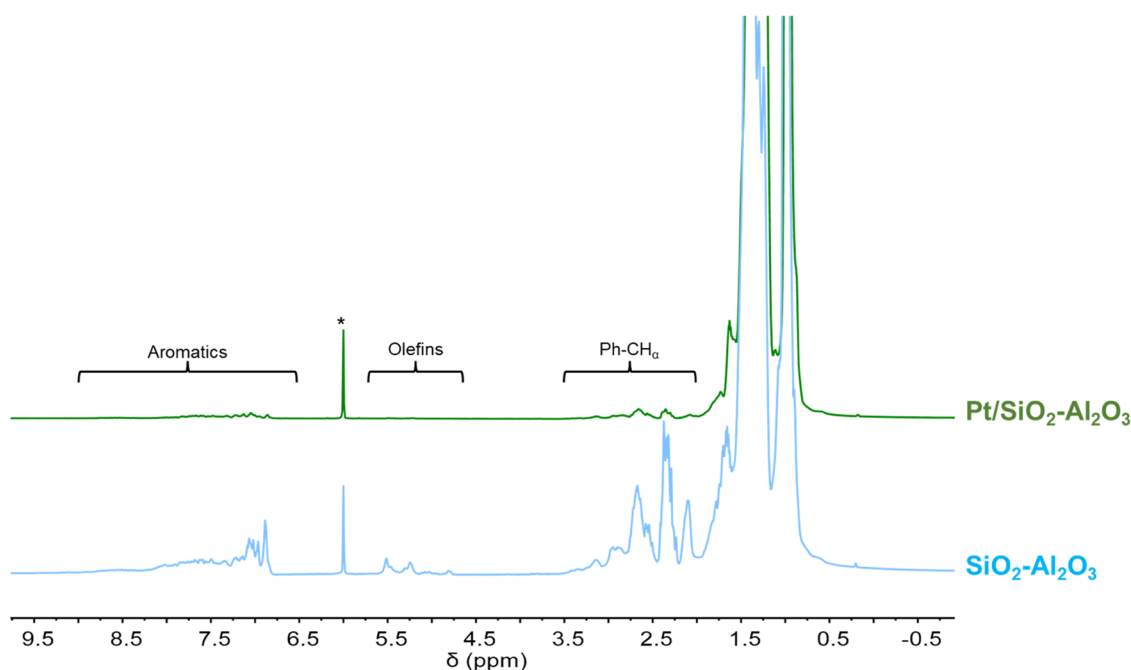


Figure 6. ^1H NMR spectra (1,1,2,2-tetrachloroethane- d_2 (*), 120 $^\circ\text{C}$, 500 MHz, concentration ≈ 0.15 M) of the liquid/wax product of PE ($M_w = \sim 3,800$ g/mol, $M_n = \sim 1,100$ g/mol) hydrogenolysis using Pt/SiO₂-Al₂O₃ (4 mol % aromatics) (top) and SiO₂-Al₂O₃ (25 mol % aromatics) (bottom). Hydrogenolysis conditions: PE (3.00 g), Pt/M_xO_y (equivalent to 4 mg of Pt), 300 $^\circ\text{C}$, 24 h, and 170 psi H₂.

Table 4. Structural Properties of the Polymer Samples after Catalytic Hydrogenolysis with Varying H₂ Pressure^a

catalyst	H ₂ (psi)	M_w^b (g/mol)	M_n^b (g/mol)	M_w/M_n^b	$N_{\text{branch}/100\text{C}}^c$	Ph-CH ₂ /H _{aromatic} ^c
Pt/SiO ₂ -Al ₂ O ₃	0	330	270	1.2	2.5	1.5
	85	360	300	1.2	6.0	1.2
	170	190	130	1.5	5.0	1.5
SiO ₂ -Al ₂ O ₃	0	260	210	1.2	4.0	1.5
	85	330	270	1.2	5.5	1.9
	170	290	250	1.2	5.5	1.8

^aHydrogenolysis conditions: PE (3.00 g; $M_w = \sim 3800$ g/mol, $M_n = \sim 1100$ g/mol, $M_w/M_n = \sim 3.45$, $N_{\text{branch}/100\text{C}} = 1.0$), Pt/M_xO_y (equivalent to 4 mg of Pt), or equivalent quantity of M_xO_y, 300 $^\circ\text{C}$, and 24 h. ^bDetermined by high-temperature GPC with RI detection. ^cDetermined by ^1H NMR spectroscopy (see the Supporting Information).

observed for Pt/SiO₂-Al₂O₃ and SiO₂-Al₂O₃ ($N_{\text{branch}/100\text{C}} = 5.5$ and 5, respectively) indicates that the increase in branching observed with Pt/SiO₂-Al₂O₃ is likely the results of hydrocracking occurring on the SiO₂-Al₂O₃ support. SiO₂-Al₂O₃ also shows increased selectivity toward aromatics compounds compared to Pt/SiO₂-Al₂O₃, forming 75 mol % alkane and alkene products and 25 mol % aromatics (16 mol % alkylbenzenes + 5 mol % alkylnaphthalenes + 4 mol % alkylphenanthrenes) in the liquid/wax products, compared to only 4 mol % aromatics for Pt/SiO₂-Al₂O₃ (Figure 6). This is expected as the SiO₂-Al₂O₃ support only contains acidic cracking sites, with no metal sites available to facilitate the hydrogenation of olefinic intermediates. This leads to increased amounts of olefins that can cyclize into aromatics, catalyzed by the acid sites of the support.⁵⁹ The formation of primarily monoalkyl aromatic products using SiO₂-Al₂O₃ (97 wt % Si) is in contrast to Pennel et al., who observed increasing polyaromatic formation with increasing silica content (1.5–30 wt % Si) for polyethylene deconstruction using SiO₂-Al₂O₃ under H₂-free conditions.⁵¹ Analysis by RT-GPC-UV showed molecular weights similar to HT-GPC-RI analysis ($M_w = 330$ g/mol and $M_n = 250$ g/mol), indicating that the UV-active aromatic chromophores are evenly distributed across the

molecular weight range (Table S8). The liquid and gas products also show increased olefin formation compared to Pt/SiO₂-Al₂O₃ (<15 and <1 mol %, respectively, for the liquid products) due to the hindered ability of the acidic support to hydrogenate the hydrocracked products.

The background reactivity of SiO₂-Al₂O₃ helps explain the higher activity observed using Pt/SiO₂-Al₂O₃ compared to Pt/SrTiO₃. Based on the similar nanoparticle diameter of these two catalysts (≈ 1 nm) and the higher active metallic surface area of Pt/SrTiO₃ compared to Pt/SiO₂-Al₂O₃ (0.95 and 0.48 m², respectively), Pt/SrTiO₃ would be expected to have a higher activity than Pt/SiO₂-Al₂O₃. However, the background reactivity and ability of SiO₂-Al₂O₃ to crack polyolefins without an active metal center pushes the activity of Pt/SiO₂-Al₂O₃ above that of Pt/SrTiO₃ (94 and 79% total conversion, respectively).

2.2.7. H₂ Dependence Using Pt/SiO₂-Al₂O₃ and SiO₂-Al₂O₃. Polyethylene hydrogenolysis reactions were also conducted using Pt/SiO₂-Al₂O₃ and SiO₂-Al₂O₃ under the same conditions with variable H₂ pressure (0, 85, and 170 psi H₂) (Table 4 and Figures 7, S28, S56–S67, and S81). The reactivity of Pt/SiO₂-Al₂O₃ shows a dependence on H₂, with conversions to liquid/wax products increasing with increasing

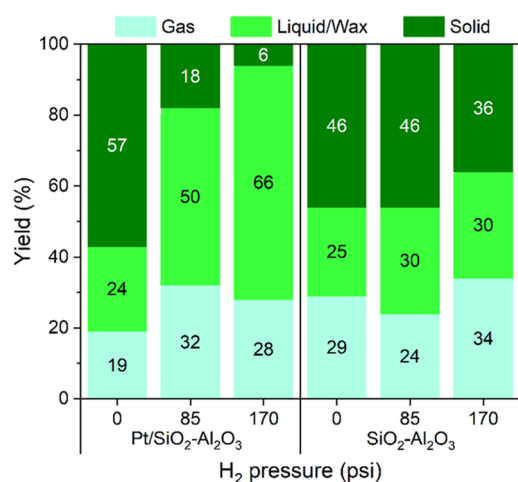


Figure 7. Gas, liquid/wax, and solid yield for the hydrogenolysis of PE ($M_w \sim 3800$ g/mol, $M_n \sim 1100$ g/mol) using Pt/SiO₂-Al₂O₃ or SiO₂-Al₂O₃ with varying H₂ pressure. Hydrogenolysis conditions: PE (3.00 g), Pt/M_xO_y (equivalent to 4 mg of Pt), or equivalent quantity of M_xO_y, 300 °C, and 24 h. Note: for reactions repeated in duplicate, the standard deviation is approximately $\pm 5\%$ yield.

H₂ pressure. In contrast, SiO₂-Al₂O₃ liquid/wax conversion is independent of H₂ pressure, with gas and liquid/wax yields within experimental uncertainty. Note that reactions were repeated at least in duplicates, and the standard deviation is approximately $\pm 5\%$ yield. This is consistent with the metal sites of Pt/SiO₂-Al₂O₃, facilitating hydrogenation of the olefinic intermediates formed via hydrocracking, in addition to allowing polymer deconstruction to occur via a hydrogenolysis pathway with reaction rates that increase with increasing H₂ availability.

2.2.8. Alternative Polyethylene Feedstocks with Pt/SiO₂-Al₂O₃. Due to the enhanced conversion of PE ($M_w \sim 3,800$ g/mol; $M_n \sim 1,100$ g/mol) to liquid products using Pt/SiO₂-Al₂O₃, further exploration of two other polyethylene substrates (Dow Affinity: $M_w = 16,400$ g/mol, $M_n = 8,700$ g/mol; Dow Engage: $M_w = 91,400$ g/mol, $M_n = 43,700$ g/mol; Tables 5 and 6 and Figures S15–S18), and a 1:1:1 mixture of the three polyethylene feedstocks (polyolefin (PO) mixture) were investigated under hydrogenolysis conditions (170 psi H₂, 300 °C, 24 h).

The gas products of the reactions show similar trends, with the major products being either methane or propane (Figure S29). The increased methane production for the PO mixture implies increased rates of hydrogenolysis compared to hydrocracking.³² The liquid/wax hydrogenolysis product was found to be independent of the polyethylene feedstock, with all

Table 6. Polymer Samples after Catalytic Hydrogenolysis with Corresponding Structural Properties^a

polymer	UPLC ^b			¹³ C{ ¹ H}
	M_w (g/mol)	M_n (g/mol)	M_w/M_n	
PE	320	280	1.1	5.5
Dow affinity	310	280	1.1	6.2
Dow engage	320	290	1.1	5.3
PO mixture ^d	320	290	1.1	6.0

^aHydrogenolysis conditions: polyethylene (3.00 g), Pt/SiO₂-Al₂O₃ (equivalent to 4 mg of Pt), 300 °C, 24 h, and 170 psi H₂.

^bDetermined by UPLC in the hexane mobile phase. ^cDetermined by ¹³C{¹H} NMR spectroscopy. ^dCorresponds to a 1:1:1 mixture of PE, Dow Affinity, and Dow Engage.

polyethylene samples forming indistinguishable products by UPLC and ¹H and ¹³C{¹H} NMR spectroscopy in similar yields (Tables 5, 6, and S9 and Figures S82 and S20). UPLC showed molecular weights within the error of analysis ($M_w = 310$ – 320 g/mol and $M_n = 280$ – 290 g/mol), with NMR spectroscopy confirming similar branch densities ($N_{\text{branch}/100\text{C}} = 5.0$ – 6.2) and branch identities (C₁–C₄ and *sec*-butyl branches; Figures S46–S48 and S68–S76). In addition, all products showed the presence of aromatics by ¹H NMR spectroscopy, with Ph-CH_α/H_{aromatic} = 1.0–1.5 and increased intensity in the -CH₂R region, indicating that the major species are, on average, di- or trialkylbenzenes with primarily -CH₂R substituents. The independence of hydrogenolysis products on polyethylene feedstock allows the potential to upcycle a variety of polyethylene mixtures with varying molecular weights and branch densities into a single product.

3. CONCLUSIONS

Herein we report a comprehensive study using platinum grafted onto six M_xO_y supports under consistent polyethylene deconstruction conditions ($T = 300$ °C, $P(\text{H}_2) = 170$ psi, $t = 24$ h). Pt/SrTiO₃ and Pt/SiO₂-Al₂O₃ catalysts gave the highest yield of liquid/wax products with significantly reduced molecular weights compared to the starting material (from $M_n < 1,100$ g/mol to $M_n < 400$ g/mol). Several key properties of the Pt/SiO₂-Al₂O₃ polymer deconstruction products indicate a primarily hydrocracking deconstruction pathway, namely, the increased production of C₃ and C₄ alkane gases compared to methane, the significantly increased number and type of branches (from $N_{\text{branch}/100\text{C}} = 1.0$ to 5.5), the presence of partially hydrogenated hydrocarbons (<1 mol %) and aromatics (4 mol %), and the dependence of catalytic activity on H₂ pressure. In contrast, and consistent with previous reports,^{8,18,20,39} the liquid/wax products produced using Pt/

Table 5. Polymer Samples with Corresponding Structural Properties and Catalytic Hydrogenolysis Product Yields^a

polymer	starting materials				product yield		
	M_w^b (g/mol)	M_n^b (g/mol)	M_w/M_n^b	$N_{\text{branch}/100\text{C}}^c$	gas ^d (%)	liquid/wax ^e (%)	solid (%)
PE	3760	1110	3.4	1.0	28	66	6
Dow affinity	16,430	8670	1.9	3.8	27	68	5
Dow engage	91,410	43,660	2.1	3.8	27	61	12
PO mixture ^f					28	68	4

^aHydrogenolysis conditions: polyethylene (3.00 g), Pt/SiO₂-Al₂O₃ (equivalent to 4 mg of Pt), 300 °C, 24 h, and 170 psi H₂. ^bDetermined by high-temperature GPC with RI detection. ^cDetermined by ¹³C{¹H} NMR spectroscopy. ^dDetermined by mass balance: gas yield = 100 – liquid/wax yield – solid yield. ^eDefined as the weight (g) of hydrocarbons recovered relative to the initial mass of PE. ^fCorresponds to a 1:1:1 mixture of PE, Dow Affinity, and Dow Engage.

SrTiO₃ showed properties consistent with a purely hydrogenolysis mechanism, such as high amounts of methane gas production, decreased branch density (from $N_{\text{branch}}/100\text{C} = 1.0$ to 0.1), and fully hydrogenated products. These mechanistic differences likely lie in the properties of the catalysts, where the lack of acidity and increased stability of Pt/SrTiO₃ facilitates polymer deconstruction via a hydrogenolysis pathway, generating products that have the capability to be used directly as lubricants, and the higher acidity and lower stability of Pt/SiO₂-Al₂O₃ enables polymer deconstruction via competing hydrogenolysis and hydrocracking pathways, generating alkane, alkene, and aromatic product mixtures that would require separation prior to further use. In addition, Pt/SrTiO₃ shows enhanced recyclability compared to Pt/SiO₂-Al₂O₃, which makes Pt/SrTiO₃ a potential candidate for commercial applications. Overall, this study highlights the importance of designing stable catalysts for plastic upcycling applications.

Investigations into the SiO₂-Al₂O₃ support as a hydrogenolysis catalyst revealed its capability to deconstruct polyethylene via a cationic cracking mechanism. Varying H₂ pressure highlighted a dependence on H₂ for Pt/SiO₂-Al₂O₃, with liquid product yield increasing with an increase in H₂ pressure (~25% with no H₂ to ~66% with 170 psi H₂) and showed no dependence on H₂ pressure for SiO₂-Al₂O₃ (liquid yield of ~30%). This is consistent with polymer deconstruction using SiO₂-Al₂O₃ proceeding via a cracking pathway, with the metal sites of Pt/SiO₂-Al₂O₃ enabling hydrogenation of the olefin products and facilitating a competing hydrogenolysis mechanism. The exploration of other SiO₂-Al₂O₃ supports with different Al contents is the focus of future work, including characterizing the effects of the metal/acid balance of these catalysts.

Finally, the products of the hydrogenolysis reactions were found to be independent of polyethylene feedstock, allowing the potential to upcycle polyethylenes with varying properties into a single product. The investigation of other polyolefins (for example, polypropylene and block copolymers) is the focus of future work.

4. EXPERIMENTAL SECTION

4.1. Commercial Reagents. M_xO_y supports, SBA-15 (Sigma-Aldrich-Millipore; mesoporous silica, particle size <150 μm, pore size 10 nm, hexagonal pore morphology; SiO₂), γ-alumina (Strem Chemicals Inc.; min. 97%, pore volume 0.43 cm³/g; Al₂O₃), mesostructured MCM-41 aluminosilicate (Sigma-Aldrich-Millipore; ~3% aluminum, pore volume 0.8–1.1 cm³/g, pore size 2–4 nm; SiO₂-Al₂O₃), cerium(IV) oxide (Sigma-Aldrich-Millipore; particle size <25 nm; CeO₂), and titanium(IV) oxide (Sigma-Aldrich-Millipore; 99.7% trace metals basis, particle size <25 nm; TiO₂) were used as received. Polyethylene (Sigma-Aldrich-Millipore, average $M_w = \sim 3,800$ g/mol and $M_n = \sim 1,100$ g/mol), linear low-density polyethylenes (Dow Chemical Company, 13 mol % 1-octene), 1,1,2,2-tetrachloroethane-*d*₂ (Cambridge Isotope Laboratories, Inc.; 99.5%), pyridine (anhydrous, 99.8%, Sigma-Aldrich), and potassium bromide (FT-IR grade, ≥99% trace metals basis, Sigma-Aldrich) were used as received.

4.2. Support Synthesis and Pretreatment. The procedure for the synthesis and pretreatment of SrTiO₃ has been previously reported by McCullough et al.³⁷ and Peczak et al.³⁹ After synthesis, SrTiO₃ was calcined at 550 °C for 4 h, treated with ozone in oxygen (8% O₃) at 200 °C for 1.5 h, treated with steam at 200 °C for 2 h, and vacuum-dried at 200 °C for 12 h. Prior to use, SiO₂-Al₂O₃, SiO₂, Al₂O₃, TiO₂, and CeO₂ were degassed at 200 °C for 12 h.

4.3. Catalyst Synthesis. The M_xO_y supports were suspended in a solution of platinum(II)acetylacetonate (Pt(acac)₃) in toluene at 80

°C for 72 h under an inert atmosphere in a N₂-filled glovebox. The samples were then washed with toluene and pentane, vacuum-dried at 60 °C overnight, calcined at 300 °C in air for 4 h, and reduced under 10% hydrogen at 300 °C for 4 h. SiO₂, Al₂O₃, Pt/SiO₂-Al₂O₃, TiO₂, and CeO₂ were then vacuum-dried at 200 °C for 12 h, while Pt/SrTiO₃ underwent another metalation cycle before drying.

4.4. Catalytic Hydrogenolysis. Hydrogenolysis experiments were performed in 100 mL Parr autoclaves (Series 4590 Micro Reactor) equipped with an overhead stirrer and thermocouple. Then, 3.00 g of polyethylene (Sigma-Aldrich, $M_w \sim 3800$ g/mol, $M_n \sim 1,100$ g/mol) and an amount of Pt/M_xO_y corresponding to 4 mg of Pt (calculated using ICP-OES wt %; polymer:catalyst ≈10:1) were placed in a glass sleeve inside the reactor. For the M_xO_y control reactions, the amount of support used was equal to the amount of support used in the Pt/M_xO_y experiments (for example, 0.233 g of Pt/SiO₂-Al₂O₃ and 0.229 g of SiO₂-Al₂O₃ for the corresponding control reaction). The autoclave was sealed and flushed with Ar and ultrahigh purity (UHP) H₂ 10 times each. The reactor was heated to 300 °C, agitated with an impeller at 800 rpm, and charged with H₂ to the desired pressure. No remarkable pressure increase was observed while heating to the desired temperature. After 24 h, the reactor was cooled to room temperature and vented to the atmosphere. The waxy product between the walls of the reactor and the walls of the glass sleeve was collected, filtered hot through silica gel (Sigma-Aldrich-Millipore; Davisil grade 646), washed with *n*-hexane, and dried under vacuum at ~55 °C. The solid remaining in the glass sleeve was washed with dichloromethane, and the filtrate dried under vacuum, typically resulting in <5 mg of soluble product, with the small quantity hindering further analysis. The gaseous yield was calculated as a mass balance considering the solid and liquid/wax yields and the mass of the starting polyethylene, which may lead to inaccuracies as mass lost during workup might be incorrectly attributed to gases. Note that reactions were repeated at least in duplicates and the standard deviation was approximately ±5% yield.

4.5. Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES). Inductively coupled plasma optical emission spectroscopy (ICP-OES) was performed at the Quantitative Bioelement Imaging Center (QBIC) at Northwestern University on a Thermo iCAP 7600 Spectrometer (Thermo Fisher, Waltham, MA).

4.6. N₂ Sorption and Brunauer–Emmett–Teller (BET) Analysis. The catalyst surface area was determined by nitrogen adsorption using a Micromeritics ASAP 2010 instrument at −196 °C. Prior to surface area measurement, the samples were degassed by heating at 200 °C under vacuum for 12 h. Seven data points were measured for an accurate evaluation of the surface area.

4.7. Powder X-ray Diffraction (PXRD). Powder X-ray diffraction (PXRD) patterns were collected on a Bruker Advance D8 powder X-ray diffractometer with Ni-filtered Cu Kα radiation operating at 40 kV and 40 mA. The scan range was between 10 and 60° with 0.006° steps at a collection speed of 1 s/step.

4.8. Transmission Electron Microscopy (TEM). Nanoparticle imaging was conducted by transmission electron microscopy using an FEI Talos F200X TEM/STEM operated at 200 kV in the Center for Nanoscale Materials (CNM) at Argonne National Laboratory. In preparation for analysis, powdered samples (~20 mg) were sonicated in ethanol (10 mL) for 15 min. The resulting suspension was drop cast onto a lacey carbon TEM grid (Ted Pella, Inc., UC-A on Lacey 400 mesh Cu).

4.9. CO Chemisorption. CO pulse chemisorption measurements were carried out by using a Micromeritics AutoChem II instrument. Prior to the measurements, all samples were reduced in 4% H₂ in balance with Ar (Airgas, certified) flowing at 40 sccm at 400 °C for 1 h. CO pulse chemisorption measurements were performed at 40 °C by using 5% CO in balance with He (Airgas, certified). In a typical experiment, 20 mg of sample was placed in a U-shaped reactor tube and dosed with dilute CO up to 20 pulses. A built-in TCD detector was employed to quantify pulse uptake, and an in-line atmospheric mass spectrometer was used to monitor the effluent gas for CO₂ production. No CO₂ was detected in any of the reported

measurements, indicating that only CO titration took place on metallic Pt surfaces.

4.10. H₂ Temperature-Programmed Reduction (H₂-TPR). H₂ temperature-programmed reduction (H₂-TPR) was performed on a home-built fixed-bed reactor connected with an online MS spectrometer (Agilent 5975 equipped with an inert Mass Selective Detector). For each trial, the sample was pretreated in He gas flow (30 mL/min) at 300 °C for 1 h to clean its surface. After cooling to room temperature in a He flow, a blank run with temperature ramping from 30 to 330 °C at a heating rate of 10 °C/min was conducted to confirm that there is no background signal. After cooling to room temperature in He again, 10 vol % H₂/He was switched into the reactor at a flow rate of 100 mL/min until a stable baseline appeared before ramping temperature. The TPR profiles (H₂O signal of $m/z = 18$) were recorded from 30 to 330 °C at a heating rate of 10 °C/min and then cooled down to room temperature. All of the MS intensity was normalized to a 100 mg of sample. H₂-TPR measurements were performed using Pt/Al₂O₃, Pt/SiO₂, Pt/TiO₂, and Pt/CeO₂ prior to the reduction step of the catalyst synthesis procedure (Section 4.3). Pt/SiO₂-Al₂O₃ was calcined in 20 vol % O₂/He at 300 °C for 4 h prior to the H₂-TPR experiments.

4.11. Acid Site Determination by IR Spectroscopy of Adsorbed Pyridine. Lewis and Brønsted acidities were measured by pyridine adsorption. The IR calibration curve reported by Sun et al. was also used here.⁴⁵ A sample of dried material (ca. 80 mg) was transferred to a Schlenk tube inside an Ar-filled glovebox. The Schlenk tube was sealed, removed from the glovebox, and connected to a high vacuum line (base pressure 30 mTorr). The tube was evacuated at 400 °C for 2 h and then cooled to room temperature. Excess dry, air-free pyridine vapor was introduced at room temperature and allowed to equilibrate for 15 min. The tube was then evacuated at ca. 30 mTorr for 30 min while heated at 150 °C to remove physisorbed pyridine. The tube was transferred into an Ar-filled glovebox, where ca. 10 mg (precisely weighed) of sample was pressed into a self-supporting pellet with a diameter of 5.0 mm using an ICL hand press. The spectrum of the glovebox atmosphere was used as the background. Due to difficulties in forming a self-supporting pellet for Pt/SiO₂-Al₂O₃, 82 mg (precisely weighed) of dry KBr (FT-IR grade) was ground with 18 mg (precisely weighed) of Pt/SiO₂-Al₂O₃ containing adsorbed pyridine in a mortar and pestle inside an Ar-filled glovebox. A portion of the mixture (40 mg, precisely weighed, containing ca. 7.6 mg of Pt/SiO₂-Al₂O₃) was pressed into a pellet with a diameter of 5.0 mm. The spectrum of a KBr pellet was used as the background. IR spectra were recorded using a Bruker α FT-IR spectrometer over the range of 4,000–400 cm⁻¹ at a resolution of 2 cm⁻¹, accumulating 64 scans. The absorbance was normalized by the precise sample mass. The areas of the peaks at 1,455 and 1,545 cm⁻¹ (Pt/SiO₂-Al₂O₃) and 1,438 and 1,553 cm⁻¹ (Pt/CeO₂),⁴ corresponding to pyridine interacting with Lewis acidic sites (LAS) and Brønsted acid sites (BAS), respectively, were used for quantification. Note: the calibration curve is measured for Al center LAS for Pt/CeO₂, and it is therefore assumed that the response factor for Ce is the same as Al.

4.12. Gas Chromatography with Flame Ionization Detection (GC-FID). GC-FID analysis was conducted on an Agilent 6890N with an Agilent, HP-Plot Al₂O₃-S, 25 m × 0.320 mm × 8.0 μ m capillary column, and a flame ionization detector. Samples were injected directly from the Parr reactor using automated sampling valves. The GC was calibrated for C₁–C₈ alkanes and alkenes, enabling accurate determination of the percentage of gases making up the sample. However, the procedure was not designed to quantify the total amount of gas products and could not accurately determine the total gas yield.

4.13. Gel-Permeation Chromatography (GPC). The molecular weights (M_n and M_w) of the starting polymers and hydrogenolysis products were determined by high-temperature gel-permeation chromatography (HT-GPC) on a 1260 Infinity II High-Temperature GPC System equipped with refractive index (RI) and viscometer detectors with three Agilent columns (PLgel MIXED-C, 7.5 × 300 mm², 5 μ m) and a guard column (PLgel MIXED, 7.5 × 50 mm, 5

μ m). Monodisperse polyethylene standards (PSS; 338–120,000 g/mol) were used to build the calibration curve. The eluent was 1,2,4-trichlorobenzene (TCB) containing 0.01 wt % 3,5-di-*tert*-butyl-4-hydroxytoluene (BHT) at a flow rate of 1.0 mL/min and a temperature of 150 °C. The samples were prepared in TCB at a concentration of ~1.0 to ~4.0 mg/mL and heated at 150 °C for 24 h prior to injection.

To verify the accuracy of the GPC at low molecular weights, linear alkane and alkene samples were analyzed. The linear hydrocarbons showed narrow dispersities and exhibited good agreement between actual and observed molecular weights (Table S1). The molecular weights of a series of liquid hydrogenolysis products were also more accurately determined by UPLC, which showed good agreement with GPC (Table S9). This further highlights GPC as an accurate method for the determination of the molecular weights of these species, despite the low molecular weights and increased branch density of the products leading to changes in the hydrodynamic properties of the polymer compared to the starting material. The molecular weights of the hydrogenolysis products containing aromatic chromophores were determined by room temperature gel-permeation chromatography (RT-GPC) on Waters 2690 high-performance liquid chromatography (HPLC) instrument equipped with a Waters 2410 refractive index (RI) detector and a Waters 2998 photodiode array detector (PDA) with two Agilent columns (PLgel, 5 μ m MiniMIX-D, 250 × 4.6 mm²) and a guard column (MW linear range 200–400,000 g/mol). Polystyrene standards (Agilent EasiVial kit; 200–400,000 g/mol) were used to build the calibration curve. The eluent was chloroform containing 0.25 vol % triethylamine at a flow rate of 0.35 mL/min and at room temperature.

4.14. Ultraperformance Liquid Chromatography (UPLC). A Waters Corporation UPLC (Acquity H-Class) system equipped with an ACQUITY AQT XT45 (1.7 μ m, 4.6 mm, 150 mm, effective molecular weight range: 200–5000 g/mol) and a refractive index detector was used to characterize the number-averaged and weight-averaged molecular weights (M_n and M_w) of the hydrogenolysis products. Research-grade (Purity ≥99.0%) alkane species (*n*-octane, *n*-nonane, *n*-decane, *n*-dodecane, *n*-hexadecane, *n*-eicosane, *n*-octacosane, *n*-dotriacontane, and *n*-hexatriacontane) were individually used and prepared in 1 mL (50 mg/mL dissolved in Optima grade *n*-hexane) as the calibration standard. A calibration curve and representative chromatography trace are presented in Figure S1. The analytes were dissolved in Optima grade *n*-hexane (1 mL, 50 mg/mL), and 5 μ L was injected for each analysis.

4.15. Nuclear Magnetic Resonance (NMR) Spectroscopy. Solution-phase nuclear magnetic resonance (NMR) spectra were collected using a Bruker Avance III 500 MHz NMR spectrometer (11.7 T) at 120 °C in 1,1,2,2-tetrachloroethane-*d*₂ and referenced internally to residual solvent signals. ¹H NMR spectra (500 MHz) were recorded with 32 scans, and ¹³C{¹H} NMR spectra (125 MHz) were recorded with ~9,500 scans. Spectra were analyzed using MestReNova (v11.0.1, Mestrelab Research S.L.). The branching density was determined by ¹³C{¹H} NMR spectroscopy: integrals were totaled to 100C, and the values for the $\alpha\beta$ -resonances used to determine branch density (37.0, 33.5, 34.3, and 34.0 for C₁, C₂, C₃, and C₄ branches, respectively).⁶⁰

4.16. Estimation of Aromatic Selectivity. Three types of aromatic species can be distinguished in the ¹H NMR spectra: 6.5–7.5 ppm (*H*₁, primarily alkylbenzenes), 7.5–8.3 ppm (*H*₂, primarily alkyl-naphthalenes), and 8.3–9.0 ppm (*H*₃, primarily alkylphenanthrenes). Using the method developed by Sun et al.⁴⁵ and Lee et al.,⁵² the selectivities (*S_i*) for each aromatic structure type were evaluated using the following equations, with the assumption that alkanes and aromatics were the only products:

$$\begin{aligned} S_{\text{phe}}: S_{\text{nap}}: S_{\text{ben}} &= n_{\text{phe}}: n_{\text{nap}}: n_{\text{ben}} \\ &= 1/2H_3: 1/4(H_2 - 3/2H_3): 1/3H_1 - 1/12H_2 \\ &\quad - 5/24H_3 \end{aligned}$$

$$S_{\text{phe}} + S_{\text{nap}} + S_{\text{ben}} + S_{\text{alk}} = 1$$

$$\frac{H_{\text{arom}}}{H_{\text{total}}} = (3S_{\text{ben}} + 5S_{\text{nap}} + 7S_{\text{phe}})/[(2\bar{i} - 6)S_{\text{ben}} + (2\bar{i} - 12)S_{\text{nap}} + (2\bar{i} - 18)S_{\text{phe}} + (2\bar{i} + 2)S_{\text{alk}}]$$

where the ratio of $H_1:H_2:H_3$ and $H_{\text{arom}}/H_{\text{total}}$ were obtained by integrating the relevant regions in the ^1H NMR spectrum, S_i corresponds to the fractional selectivities for each aromatic product (S_{ben} = alkylbenzenes, S_{nap} = alkylnaphthalenes, S_{phe} = alkylphenanthrenes, and S_{alk} = alkanes), n_i corresponds to the molar amount for each aromatic structure type, and $\bar{i}$ corresponds to the average carbon number, determined by GPC.

4.17. Estimation of Olefin Content. Samples were dissolved in $\text{TCE-}d_2$ with a known concentration of 1,2-dichloroethane (DCE) as an internal standard. The olefin content was determined based on the integration of the olefin resonances at 4.60–5.60 ppm relative to the integration of the internal standard resonance at 3.79 ppm. The moles of the hydrogenolysis liquid/wax sample added to the NMR spectroscopy tube were determined using m/M_n , with the M_n value obtained from GPC.

4.18. Average Rate of Carbon–Carbon Bond Scission and Turnover Frequency. The average rate of C–C bond scission ($r_{\text{C-C}}$) was calculated using a method similar to that reported by Sun et al.⁴⁵ using the following equations:

$$N(t) = \sum_1^8 n_{\text{gas}} + \frac{m_{\text{liq}}}{M_n(\text{liq})} + \frac{m_{\text{solid}}}{M_n(\text{PE})}$$

$$N(0) = \frac{m_{\text{PE}}}{M_n(\text{PE})}$$

$$r_{\text{C-C}} = \frac{N(t) - N(0)}{t}$$

where $\sum_1^8 n_{\text{gas}}$ refers to the sum of moles of $\text{C}_1\text{--C}_8$, m_{liq} refers to the mass of liquid products, $M_n(\text{liq})$ refers to the M_n of the liquid products, m_{solid} refers to the mass of the solid residue, $M_n(\text{PE})$ refers to the M_n of the polyethylene starting material, and m_{PE} refers to the mass of the initial polyethylene, respectively.

In this calculation of $r_{\text{C-C}}$, n_{gas} carries some error as the overall gas yield is determined by mass balance considering the liquid/wax yield, solid yield, and starting mass of polyethylene. Using these values, the average turnover frequency (TOF) for Pt/SrTiO₃ was determined using the equation below by converting the metallic surface area of Pt, obtained from CO chemisorption, into mmol of surface Pt. The cross-sectional area of the Pt atoms was taken to be 0.08 nm.

$$\text{TOF (s}^{-1}\text{)} = \left(\frac{r_{\text{C-C}} (\text{mmol} \cdot \text{h}^{-1})}{\text{Pt (mmol)}} \right) / 3600$$

Average TOF is determined as product analysis is only performed upon completion of the reaction. Therefore, an initial rate measurement is not captured, which may decrease as the reaction proceeds. Note also that the turnover frequency (TOF) for Pt/SiO₂–Al₂O₃ cannot be calculated because Pt is not solely responsible for the reaction rate (acid site catalyzed hydrocracking).

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.3c15350>.

Additional experimental details; catalyst characterization data (ICP-OES, PXRD patterns, STEM images, CO chemisorption, BET analysis, H₂-TPR profiles, and acid site determination); polyethylene starting material characterization (NMR spectra); and PE hydrogenolysis

product analysis (yield information, GC-FID data, GPC chromatograms, and NMR spectra). (PDF)

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J.V.L., M.M.A.-O., and M.D. designed the experiments. J.V.L. carried out the bulk of the laboratory research and wrote the draft of the manuscript. Y.-H.L. carried out NMR and GPC analysis of aromatic hydrogenolysis products. J.S. carried out acidity analysis of the catalysts. C.B. carried out TEM analysis of the catalysts. R.U. carried out chemisorption analysis of the catalysts. R.M.K. performed CO chemisorption analysis. C.M. carried out H₂-TPR analysis of the catalysts. R.K.B. carried out H₂-TPR analysis of the catalysts. Y.-Y.W. carried out UPLC analysis. L.Q. developed the UPLC method. M.S.F. carried out the synthesis of the Pt/STO catalyst. A.D.S., W.H., S.L.S., K.R.P., M.M.A.-O., and M.D. obtained the funding and supervised the work. This manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare the following competing financial interest(s): Dr. Robert Kennedy is the CTO of Aeternal Upcycling which commercialize waste plastic to liquid products technology.

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REFERENCES

- (1) Ritchie, H.; Roser, M. Our World in Data, Plastic Pollution 2023 <https://ourworldindata.org/plastic-pollution>.
- (2) Executive Summary OECD: 2022.
- (3) Plastics Flows and their Impacts on the Environment OECD: 2022.

- (4) Li, H.; Aguirre-Villegas, H. A.; Allen, R. D.; Bai, X.; Benson, C. H.; Beckham, G. T.; Bradshaw, S. L.; Brown, J. L.; Brown, R. C.; Cecon, V. S.; Curley, J. B.; Curtzwiler, G. W.; Dong, S.; Gaddameedi, S.; Garcia, J. E.; Hermans, I.; Kim, M. S.; Ma, J.; Mark, L. O.; Mavrikakis, M.; Olafasakin, O. O.; Osswald, T. A.; Papanikolaou, K. G.; Radhakrishnan, H.; Sanchez Castillo, M. A.; Sánchez-Rivera, K. L.; Tumu, K. N.; Van Lehn, R. C.; Vorst, K. L.; Wright, M. M.; Wu, J.; Zavala, V. M.; Zhou, P.; Huber, G. W. Expanding Plastics Recycling Technologies: Chemical Aspects, Technology Status and Challenges. *Green Chem.* **2022**, *24* (23), 8899–9002.
- (5) Jia, X.; Qin, C.; Friedberger, T.; Guan, Z.; Huang, Z. Efficient and Selective Degradation of Polyethylenes into Liquid Fuels and Waxes Under Mild Conditions. *Sci. Adv.* **2016**, *2* (6), No. e1501591.
- (6) Liu, S.; Kots, P. A.; Vance, B. C.; Danielson, A.; Vlachos, D. G. Plastic Waste to Fuels by Hydrocracking at Mild Conditions. *Sci. Adv.* **2021**, *7* (17), No. eabf8283.
- (7) Tennakoon, A.; Wu, X.; Paterson, A. L.; Patnaik, S.; Pei, Y.; LaPointe, A. M.; Ammal, S. C.; Hackler, R. A.; Heyden, A.; Slowing, I. I.; Coates, G. W.; Delferro, M.; Peters, B.; Huang, W.; Sadow, A. D.; Perras, F. A. Catalytic Upcycling of High-Density Polyethylene via a Processive Mechanism. *Nat. Catal.* **2020**, *3* (11), 893–901.
- (8) Hackler, R. A.; Vyavhare, K.; Kennedy, R. M.; Celik, G.; Kanbur, U.; Griffin, P. J.; Sadow, A. D.; Zang, G.; Elgowainy, A.; Sun, P.; Poeppelmeier, K. R.; Erdemir, A.; Delferro, M. Synthetic Lubricants Derived from Plastic Waste and their Tribological Performance. *ChemSusChem* **2021**, *14* (19), 4181–4189.
- (9) Kots, P. A.; Liu, S.; Vance, B. C.; Wang, C.; Sheehan, J. D.; Vlachos, D. G. Polypropylene Plastic Waste Conversion to Lubricants Over Ru/TiO₂ Catalysts. *ACS Catal.* **2021**, *11* (13), 8104–8115.
- (10) Jaydev, S. D.; Martin, A. J.; Pérez-Ramírez, J. Direct Conversion of Polypropylene into Liquid Hydrocarbons on Carbon-Supported Platinum Catalysts. *ChemSusChem* **2021**, *14* (23), 5179–5185.
- (11) Kanbur, U.; Zang, G.; Paterson, A. L.; Chatterjee, P.; Hackler, R. A.; Delferro, M.; Slowing, I. I.; Perras, F. A.; Sun, P.; Sadow, A. D. Catalytic Carbon-Carbon Bond Cleavage and Carbon-Element Bond Formation Give New Life for Polyolefins as Biodegradable Surfactants. *Chem* **2021**, *7* (5), 1347–1362.
- (12) Arroyave, A.; Cui, S.; Lopez, J. C.; Kocen, A. L.; LaPointe, A. M.; Delferro, M.; Coates, G. W. Catalytic Chemical Recycling of Post-Consumer Polyethylene. *J. Am. Chem. Soc.* **2022**, *144* (51), 23280–23285.
- (13) Chen, L.; Malollari, K. G.; Uliana, A.; Sanchez, D.; Messersmith, P. B.; Hartwig, J. F. Selective, Catalytic Oxidations of C–H Bonds in Polyethylenes Produce Functional Materials with Enhanced Adhesion. *Chem* **2021**, *7* (1), 137–145.
- (14) Liu, D.; Bielawski, C. W. Post-Polymerization Modification of Poly(Vinyl Ether)s: A Ru-Catalyzed Oxidative Synthesis of Poly(Vinyl Ester)s and Poly(Propenyl Ester)s. *Polym. Chem.* **2016**, *7* (1), 63–68.
- (15) Bunesco, A.; Lee, S.; Li, Q.; Hartwig, J. F. Catalytic Hydroxylation of Polyethylenes. *ACS Cent. Sci.* **2017**, *3* (8), 895–903.
- (16) De Andrés, J.; Aguilar, A.; Domenech, J. Kinetic Study of the Catalytic Effect of Zinc(II) Ion on the Dehydrogenation of Atactic Polypropylene by Sulfur. *J. Macromol. Sci., Part A* **1990**, *27* (2), 213–223.
- (17) Ray, A.; Zhu, K.; Kissin, Y. V.; Cherian, A. E.; Coates, G. W.; Goldman, A. S. Dehydrogenation of Aliphatic Polyolefins Catalyzed by Pincer-Ligated Iridium Complexes. *Chem. Commun.* **2005**, No. 27, 3388–3390.
- (18) Celik, G.; Kennedy, R. M.; Hackler, R. A.; Ferrandon, M.; Tennakoon, A.; Patnaik, S.; LaPointe, A. M.; Ammal, S. C.; Heyden, A.; Perras, F. A.; Pruski, M.; Scott, S. L.; Poeppelmeier, K. R.; Sadow, A. D.; Delferro, M. Upcycling Single-Use Polyethylene into High-Quality Liquid Products. *ACS Cent. Sci.* **2019**, *5* (11), 1795–1803.
- (19) Zhang, F.; Zeng, M.; Yappert, R. D.; Sun, J.; Lee, Y.-H.; LaPointe, A. M.; Peters, B.; Abu-Omar, M. M.; Scott, S. L. Polyethylene Upcycling to Long-Chain Alkylaromatics by Tandem Hydrogenolysis/Aromatization. *Science* **2020**, *370* (6515), 437–441.

- (20) Hackler, R. A.; Lamb, J. V.; Peczak, I. L.; Kennedy, R. M.; Kanbur, U.; LaPointe, A. M.; Poeppelmeier, K. R.; Sadow, A. D.; Delferro, M. Effect of Macro- and Microstructures on Catalytic Hydrogenolysis of Polyolefins. *Macromolecules* **2022**, *55* (15), 6801–6810.
- (21) Rorrer, J. E.; Beckham, G. T.; Román-Leshkov, Y. Conversion of Polyolefin Waste to Liquid Alkanes with Ru-Based Catalysts Under Mild Conditions. *JACS Au* **2021**, *1* (1), 8–12.
- (22) Faust, K.; Denifl, P.; Hapke, M. Recent Advances in Catalytic Chemical Recycling of Polyolefins. *ChemCatChem* **2023**, *15*, No. e202300310.
- (23) Chen, L.; Zhu, Y.; Meyer, L. C.; Hale, L. V.; Le, T. T.; Karkamkar, A.; Lercher, J. A.; Gutiérrez, O. Y.; Szanyi, J. Effect of Reaction Conditions on the Hydrogenolysis of Polypropylene and Polyethylene into Gas and Liquid Alkanes. *React. Chem. Eng.* **2022**, *7* (4), 844–854.
- (24) Jia, C.; Xie, S.; Zhang, W.; Intan, N. N.; Sampath, J.; Pfaendner, J.; Lin, H. Deconstruction of High-Density Polyethylene into Liquid Hydrocarbon Fuels and Lubricants by Hydrogenolysis Over Ru Catalyst. *Chem. Catal.* **2021**, *1* (2), 437–455.
- (25) Rorrer, J. E.; Troyano-Valls, C.; Beckham, G. T.; Román-Leshkov, Y. Hydrogenolysis of Polypropylene and Mixed Polyolefin Plastic Waste Over Ru/C to Produce Liquid Alkanes. *ACS Sustainable Chem. Eng.* **2021**, *9* (35), 11661–11666.
- (26) Nakaji, Y.; Tamura, M.; Miyaoka, S.; Kumagai, S.; Tanji, M.; Nakagawa, Y.; Yoshioka, T.; Tomishige, K. Low-Temperature Catalytic Upgrading of Waste Polyolefinic Plastics into Liquid Fuels and Waxes. *Appl. Catal., B* **2021**, *285*, No. 119805.
- (27) Jing, Y.; Wang, Y.; Furukawa, S.; Xia, J.; Sun, C.; Hülsey, M. J.; Wang, H.; Guo, Y.; Liu, X.; Yan, N. Towards the Circular Economy: Converting Aromatic Plastic Waste Back to Arenes Over a Ru/Nb₂O₅ Catalyst. *Angew. Chem., Int. Ed.* **2021**, *60* (10), 5527–5535.
- (28) Wu, X.; Tennakoon, A.; Yappert, R.; Esveld, M.; Ferrandon, M. S.; Hackler, R. A.; LaPointe, A. M.; Heyden, A.; Delferro, M.; Peters, B.; Sadow, A. D.; Huang, W. Size-Controlled Nanoparticles Embedded in a Mesoporous Architecture Leading to Efficient and Selective Hydrogenolysis of Polyolefins. *J. Am. Chem. Soc.* **2022**, *144* (12), 5323–5334.
- (29) Kots, P. A.; Vance, B. C.; Vlachos, D. G. Polyolefin Plastic Waste Hydroconversion to Fuels, Lubricants, and Waxes: A Comparative Study. *React. Chem. Eng.* **2021**, *7* (1), 41–54.
- (30) Peczak, I. L.; Kennedy, R. M.; Hackler, R. A.; Lee, B.; Meirow, M.; Luijten, E.; Poeppelmeier, K. R.; Delferro, M. Treasuring Trash: Pt/SrTiO₃ Catalysts Process Plastic Waste into High-Value Materials. *Matter* **2023**, *6* (10), 3296–3321.
- (31) Vance, B. C.; Kots, P. A.; Wang, C.; Hinton, Z. R.; Quinn, C. M.; Epps, T. H.; Korley, L. T. J.; Vlachos, D. G. Single Pot Catalyst Strategy to Branched Products via Adhesive Isomerization and Hydrocracking of Polyethylene Over Platinum Tungstated Zirconia. *Appl. Catal., B* **2021**, *299*, No. 120483.
- (32) Lee, W.-T.; van Muyden, A.; Bobbink, F. D.; Mensi, M. D.; Carullo, J. R.; Dyson, P. J. Mechanistic Classification and Benchmarking of Polyolefin Depolymerization Over Silica-Alumina-Based Catalysts. *Nat. Commun.* **2022**, *13* (1), No. 4850.
- (33) Pines, H.; Haag, W. O. Alumina: Catalyst and Support. I. Alumina, its Intrinsic Acidity and Catalytic Activity. *J. Am. Chem. Soc.* **1960**, *82* (10), 2471–2483.
- (34) Weisz, P. B. Zeolites - New Horizons in Catalysis. *Chem. Technol.* **1973**, *3*, 498–505.
- (35) Munir, D.; Irfan, M. F.; Usman, M. R. Hydrocracking of Virgin and Waste Plastics: A Detailed Review. *Renewable Sustainable Energy Rev.* **2018**, *90*, 490–515.
- (36) Borkar, S. S.; Helmer, R.; Mahnaz, F.; Majzoub, W.; Mahmoud, W.; Al-Rawashdeh, M.; Shetty, M. Enabling Resource Circularity Through Thermo-Catalytic and Solvent-Based Conversion of Waste Plastics. *Chem. Catal.* **2022**, *2* (12), 3320–3356.
- (37) McCullough, K. E.; Peczak, I. L.; Kennedy, R. M.; Wang, Y.-Y.; Lin, J.; Wu, X.; Paterson, A. L.; Perras, F. A.; Hall, J.; Kropf, A. J.; Hackler, R. A.; Shin, Y.; Niklas, J.; Poluektov, O. G.; Wen, J.; Huang, W.; Sadow, A. D.; Poeppelmeier, K. R.; Delferro, M.; Ferrandon, M. S. Synthesis of Platinum Nanoparticles on Strontium Titanate Nanocuboids via Surface Organometallic Grafting for the Catalytic Hydrogenolysis of Plastic Waste. *J. Mater. Chem. A* **2023**, *11*, 1216–1231.
- (38) Mouat, A. R.; Whitford, C. L.; Chen, B.-R.; Liu, S.; Perras, F. A.; Pruski, M.; Bedzyk, M. J.; Delferro, M.; Stair, P. C.; Marks, T. J. Synthesis of Supported Pd⁰ Nanoparticles from a Single-Site Pd²⁺ Surface Complex By Alkene Reduction. *Chem. Mater.* **2018**, *30* (3), 1032–1044.
- (39) Peczak, I. L.; Kennedy, R. M.; Hackler, R. A.; Wang, R.; Shin, Y.; Delferro, M.; Poeppelmeier, K. R. Scalable Synthesis of Pt/SrTiO₃ Hydrogenolysis Catalysts in Pursuit of Manufacturing-Relevant Waste Plastic Solutions. *ACS Appl. Mater. Interfaces* **2021**, *13* (49), 58691–58700.
- (40) Copéret, C. Single-Sites and Nanoparticles at Tailored Interfaces Prepared via Surface Organometallic Chemistry from Thermolytic Molecular Precursors. *Acc. Chem. Res.* **2019**, *52* (6), 1697–1708.
- (41) Busca, G. Silica-Alumina Catalytic Materials: A Critical Review. *Catal. Today* **2020**, *357*, 621–629.
- (42) Rioux, R. M.; Song, H.; Hoefelmeyer, J. D.; Yang, P.; Somorjai, G. A. High-Surface-Area Catalyst Design: Synthesis, Characterization, and Reaction Studies of Platinum Nanoparticles in Mesoporous SBA-15 silica. *J. Phys. Chem. B* **2005**, *109* (6), 2192–2202.
- (43) Klasovsky, F.; Claus, P. *Metal Nanoclusters in Catalysis and Materials Science: The Issue of Size Control*; Corain, B.; Schmid, G.; Toshima, N., Eds.; Elsevier: Amsterdam, The Netherlands, 2011.
- (44) Parry, E. P. An Infrared Study of Pyridine Adsorbed on Acidic Solids. Characterization of Surface Acidity. *J. Catal.* **1963**, *2* (5), 371–379.
- (45) Sun, J.; Lee, Y.-H.; Yappert, R. D.; LaPointe, A. M.; Coates, G. W.; Peters, B.; Abu-Omar, M. M.; Scott, S. L. Bifunctional Tandem Catalytic Upcycling of Polyethylene to Surfactant-Range Alkylaromatics. *Chem* **2023**, *9* (8), 2318–2336.
- (46) Rabuffetti, F. A.; Stair, P. C.; Poeppelmeier, K. R. Synthesis-Dependent Surface Acidity and Structure of SrTiO₃ Nanoparticles. *J. Phys. Chem. C* **2010**, *114* (25), 11056–11067.
- (47) Flaherty, D. W.; Hibbitts, D. D.; Iglesia, E. Metal-Catalyzed C–C Bond Cleavage in Alkanes: Effects of Methyl Substitution on Transition-State Structures and Stability. *J. Am. Chem. Soc.* **2014**, *136* (27), 9664–9676.
- (48) Flaherty, D. W.; Iglesia, E. Transition-State Enthalpy and Entropy Effects on Reactivity and Selectivity in Hydrogenolysis of n-Alkanes. *J. Am. Chem. Soc.* **2013**, *135* (49), 18586–18599.
- (49) Santos, E.; Rijo, B.; Lemos, F.; Lemos, M. A. N. D. A. A Catalytic Reactive Distillation Approach to High Density Polyethylene Pyrolysis – Part 1 – Light Olefin Production. *Chem. Eng. J.* **2019**, *378*, No. 122077.
- (50) Ochoa, R.; Van Woert, H.; Lee, W. H.; Subramanian, R.; Kugler, E.; Eklund, P. C. Catalytic Degradation of Medium Density Polyethylene Over Silica-Alumina Supports. *Fuel Process. Technol.* **1996**, *49* (1), 119–136.
- (51) Pennel, M. L.; Maurya, A. K.; Ebrahim, A. M.; Tassone, C. J.; Cargnello, M. Activity of Silica-Alumina for the Conversion of Polyethylene into Tunable Aromatics Below Pyrolytic Temperatures. *ACS Sustainable Chem. Eng.* **2023**, *11* (34), 12623–12630.
- (52) Lee, Y.-H.; Sun, J.; Scott, S. L.; Abu-Omar, M. M. Quantitative Analyses of Products and Rates in Polyethylene Depolymerization and Upcycling. *STAR Protoc.* **2023**, *4* (4), No. 102575.
- (53) Hensen, E. J. M.; Poduval, D. G.; Degirmenci, V.; Ligthart, D. A. J. M.; Chen, W.; Maugé, F.; Rigutto, M. S.; Veen, J. A. R. v. Acidity Characterization of Amorphous Silica–Alumina. *J. Phys. Chem. C* **2012**, *116* (40), 21416–21429.
- (54) Matsushashi, H.; Nishiyama, S.; Miura, H.; Eguchi, K.; Hasegawa, K.; Iizuka, Y.; Igarashi, A.; Katada, N.; Kobayashi, J.; Kubota, T.; Mori, T.; Nakai, K.; Okazaki, N.; Sugioaka, M.; Umeki, T.; Yazawa, Y.; Lu, D. Effect of Preparation Conditions on Platinum Metal Dispersion and Turnover Frequency of Several Reactions Over

Platinum-Supported on Alumina Catalysts. *Appl. Catal., A* **2004**, 272 (1), 329–338.

(55) Enterkin, J. A.; Poeppelmeier, K. R.; Marks, L. D. Oriented Catalytic Platinum Nanoparticles on High Surface Area Strontium Titanate Nanocuboids. *Nano Lett.* **2011**, 11 (3), 993–997.

(56) Choi, I.-H.; Lee, H.-J.; Rhim, G.-B.; Chun, D.-H.; Lee, K.-H.; Hwang, K.-R. Catalytic Hydrocracking of Heavy Wax from Pyrolysis of Plastic Wastes Using Pd/H β for Naphtha-Ranged Hydrocarbon Production. *J. Anal. Appl. Pyrolysis* **2022**, 161, No. 105424.

(57) Sun, M.; Zhu, L.; Liu, W.; Zhao, X.; Zhang, Y.; Luo, H.; Miao, G.; Li, S.; Yin, S.; Kong, L. Efficient Upgrading of Polyolefin Plastics into C5–C12 Gasoline Alkanes Over a Pt/W/Beta Catalyst. *Sustainable Energy Fuels* **2022**, 6 (2), 271–275.

(58) Tan, J. Z.; Hullfish, C. W.; Zheng, Y.; Koel, B. E.; Sarazen, M. L. Conversion of Polyethylene Waste to Short Chain Hydrocarbons Under Mild Temperature and Hydrogen Pressure with Metal-Free and Metal-Loaded MFI Zeolites. *Appl. Catal., B* **2023**, 338, No. 123028.

(59) Bhan, A.; Delgass, W. N. Propane Aromatization Over HZSM-5 and Ga/HZSM-5 Catalysts. *Catal. Rev.* **2008**, 50 (1), 19–151.

(60) Wiedemann, T.; Voit, G.; Tchernook, A.; Roesle, P.; Göttker-Schnetmann, I.; Mecking, S. Monofunctional Hyperbranched Ethylene Oligomers. *J. Am. Chem. Soc.* **2014**, 136 (5), 2078–2085.